

ImPRESS Statistical Analysis Report

Baseline Review for the Newly Diagnosed Glioblastoma Cohort

ImPRESS Statistical Team

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Introduction

The ImPRESS study is a prospective investigation of losartan as an adjunctive therapy for patients with glioblastoma. According to Protocol Version 7.0 (17-Dec-2024), the trial collects comprehensive demographic, clinical, imaging, and quality-of-life information prior to treatment initiation. The statistical objectives for the interim analyses focus on characterising baseline status, neurological function, and tumour burden before randomisation to evaluate comparability across planned treatment arms.

This report summarises the currently available baseline data with emphasis on clinical status, imaging assessments, and prior treatments that may influence subsequent outcomes. The material is intended to provide early insight into data quality and cohort composition ahead of formal efficacy or safety analyses.

Randomisation Status

The present dataset is built on *sham randomisation* generated from the most recent export (see `_targets.R` workflow). No true treatment assignments have been unlocked. All findings in this document should therefore be interpreted as preliminary and descriptive only. Final efficacy comparisons will not be conducted until the database is locked and genuine randomisation codes are released.

Cohort Scope

The results herein are restricted to participants enrolled in the **newly diagnosed glioblastoma (AN) cohort**. Subjects with recurrent disease are excluded from all summaries, figures, and tables in this draft. A combined-cohort analysis will be delivered once recruitment and cleaning are complete for both strata.

Future revisions of this report will:

- incorporate the validated randomisation list once database lock is confirmed,
- integrate data from the recurrent glioblastoma cohort,
- extend the analyses to longitudinal efficacy and safety endpoints.

Baseline Characteristics

This section summarises baseline characteristics for the newly diagnosed glioblastoma cohort using the `adb1` ADaM dataset.

Demographics and Baseline characteristics

	all obs
Age (years)	
n	54
Mean (SD)	52.6 (10.2)
Median	55.0
Min - Max	26.0 - 69.0
Sex	
n	54

Demographics and Baseline characteristics

	all obs
Female	29 (53.7%)
Male	25 (46.3%)
Cohort	
n	54
Newly diagnosed glioblastoma	54 (100%)
Country	
n	54
NOR	54 (100%)
Childbearing potential?	
n	29
No	13 (44.8%)
Yes	16 (55.2%)
Diagnosis	
n	54
Newly diagnosed glioblastoma	54 (100%)
Time from surgery to randomisation (days)	
n	54
Mean (SD)	27.4 (7.3)
Median	25.5
Min - Max	13.0 - 48.0
IDH1/2 mutation	
n	54
Negative	47 (87%)
Positive	7 (13%)
MGMT status	
n	54
Negative	30 (55.6%)
Positive	24 (44.4%)
Pregnancy test result	
n	13
Negative	13 (100%)
ECOG Score	
n	54
Mean (SD)	0.3 (0.5)

Demographics and Baseline characteristics

	all obs
Median	0.0
Min - Max	0.0 - 2.0
ECOG Score (categorical)	
n	54
ECOG 0	40 (74.1%)
ECOG 1	13 (24.1%)
ECOG 2	1 (1.9%)
Karnofsky Performance Scale	
n	54
Mean (SD)	95.0 (8.0)
Median	100.0
Min - Max	60.0 - 100.0
Karnofsky Performance Scale (categorical)	
n	54
KPS 100	34 (63%)
KPS 60	1 (1.9%)
KPS 80	4 (7.4%)
KPS 90	15 (27.8%)
Total NANO score	
n	54
Mean (SD)	1.0 (1.3)
Median	0.0
Min - Max	0.0 - 4.0
EORTC QLQ-C30 score	
n	53
Mean (SD)	56.3 (8.6)
Median	54.0
Min - Max	41.0 - 77.0
EORTC QLQ-BN20 score	
n	53
Mean (SD)	30.8 (6.6)
Median	30.0
Min - Max	20.0 - 49.0
Systolic blood pressure	

Demographics and Baseline characteristics

	all obs
n	54
Mean (SD)	130.0 (15.4)
Median	128.5
Min - Max	106.0 - 180.0
Diastolic blood pressure	
n	54
Mean (SD)	81.9 (7.1)
Median	82.0
Min - Max	67.0 - 97.0
Pulse Rate	
n	54
Mean (SD)	80.7 (15.7)
Median	78.0
Min - Max	54.0 - 114.0
Weight (kg)	
n	54
Mean (SD)	78.8 (13.2)
Median	78.3
Min - Max	48.2 - 116.0
BMI	
n	51
Mean (SD)	25.0 (4.2)
Median	23.9
Min - Max	18.6 - 39.9
Height (cm)	
n	51
Mean (SD)	176.5 (8.8)
Median	175.0
Min - Max	158.0 - 194.0
Has the patient received chemotherapy	
n	54
No	54 (100%)
Has the patient received immunotherapy	
n	54

Demographics and Baseline characteristics

	all obs
No	54 (100%)
Has the patient received radiation therapy to the brain?	
n	54
No	54 (100%)
CNS surgery	
n	54
No	3 (5.6%)
Yes	51 (94.4%)
Steroids at time of MRI	
n	54
No	42 (77.8%)
Yes	12 (22.2%)
Mean relative cerebral blood flow (PTA-ROI)	
n	54
Mean (SD)	1.5 (0.4)
Median	1.4
Min - Max	0.7 - 2.4
Mean relative cerebral blood flow (ROI2)	
n	47
Mean (SD)	1.1 (0.5)
Median	1.0
Min - Max	0.2 - 2.2
Mean relative solid stress (PTA-ROI)	
n	41
Mean (SD)	0.8 (0.1)
Median	0.8
Min - Max	0.5 - 1.1
Mean relative solid stress (ROI2)	
n	32
Mean (SD)	0.9 (0.2)
Median	0.9
Min - Max	0.5 - 1.4
Mean relative apparent diffusion coefficient	
n	52

Demographics and Baseline characteristics

	all obs
Mean (SD)	1.4 (0.2)
Median	1.4
Min - Max	1.0 - 2.0
Mean relative fractional anisotropy	
n	52
Mean (SD)	0.5 (0.1)
Median	0.5
Min - Max	0.3 - 1.0
Mean relative restricted spectrum imaging	
n	50
Mean (SD)	0.5 (0.2)
Median	0.5
Min - Max	0.2 - 1.1
Mean DSC K2 leakage	
n	54
Mean (SD)	0.2 (1.6)
Median	0.4
Min - Max	-3.9 - 4.5
Mean relative cerebral blood	
n	54
Mean (SD)	1.7 (0.5)
Median	1.7
Min - Max	1.0 - 2.8
Mean relative transit time	
n	54
Mean (SD)	1.4 (0.3)
Median	1.4
Min - Max	0.9 - 2.3
Total volume of contrast agent enhancing lesion on T1-weighted MRI in cm3	
n	54
Mean (SD)	6.3 (6.5)
Median	4.3
Min - Max	0.0 - 26.3
Total volume of pathologic signal on T2/FLAIR-weighted MRI in cm3	

Demographics and Baseline characteristics

	all obs
n	54
Mean (SD)	13.5 (14.3)
Median	8.5
Min - Max	0.1 - 57.4
Mean tissue elasticity	
n	41
Mean (SD)	0.8 (0.1)
Median	0.8
Min - Max	0.5 - 1.2
Mean tissue relative viscosity	
n	41
Mean (SD)	0.9 (0.2)
Median	0.9
Min - Max	0.0 - 1.1
Mean tissue stiffness	
n	41
Mean (SD)	0.8 (0.1)
Median	0.8
Min - Max	0.5 - 1.1
Mean tissue viscosity	
n	41
Mean (SD)	0.7 (0.2)
Median	0.7
Min - Max	-0.0 - 1.1
Mean relative DSC extravasation	
n	28
Mean (SD)	1.2 (0.8)
Median	1.0
Min - Max	0.7 - 5.1
Mean relative direction value	
n	28
Mean (SD)	0.4 (1.4)
Median	0.3
Min - Max	-2.7 - 5.4

Demographics and Baseline characteristics

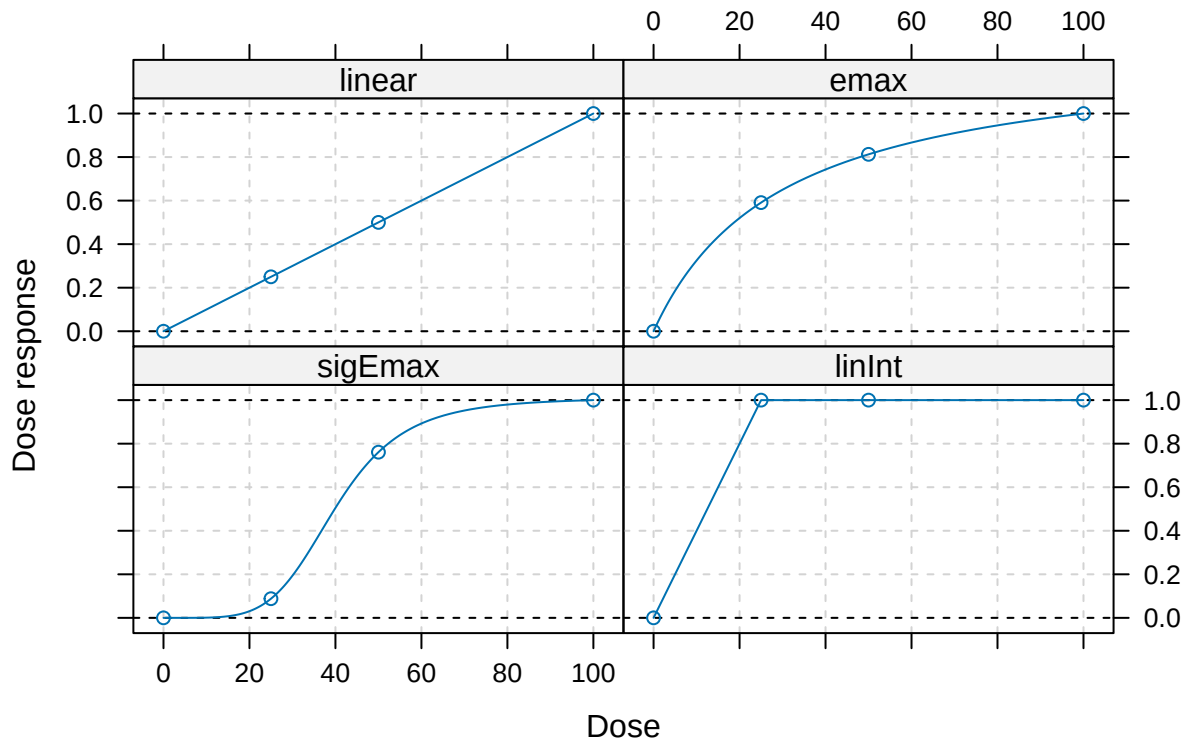
	all obs
Mean relative vessel caliber	
n	28
Mean (SD)	1.0 (0.3)
Median	1.0
Min - Max	0.5 - 1.9
Mean relative vessel size index	
n	28
Mean (SD)	1.4 (0.6)
Median	1.2
Min - Max	0.6 - 2.6
Mean relative vortex area	
n	28
Mean (SD)	1.0 (0.1)
Median	1.0
Min - Max	0.8 - 1.1

Primary Efficacy Analyses (MRI)

The primary variable is analysed with the MCP-Mod methodology as described in the protocol and SAP. We pre-specify four candidate dose-response models (linear, Emax, sigmoid Emax, logistic) and derive optimal contrasts using the estimated variance-covariance from the mixed model. The MCP step tests for a dose-response signal across candidate models with multiplicity control; if significant, the Mod step selects the best-fitting model (by AIC) and reports ED values. Results below are grouped by MRI endpoint (primary, key secondary, sensitivity).

Candidate dose-response models

Figure 1: Pre-specified candidate dose-response models



Below we summarise the MCP-Mod analyses for the four pre-specified MRI endpoints, followed by summary contrasts for the remaining MRI endpoints.

Mean relative cerebral blood flow (PTA-ROI)

Optimal contrasts

Treatment	linear	emax	sigEmax	linInt
0 mg	-0.650	-0.818	-0.595	-0.866
25 mg	-0.206	0.026	-0.382	0.284
50 mg	0.136	0.304	0.378	0.300
100 mg	0.719	0.488	0.599	0.282

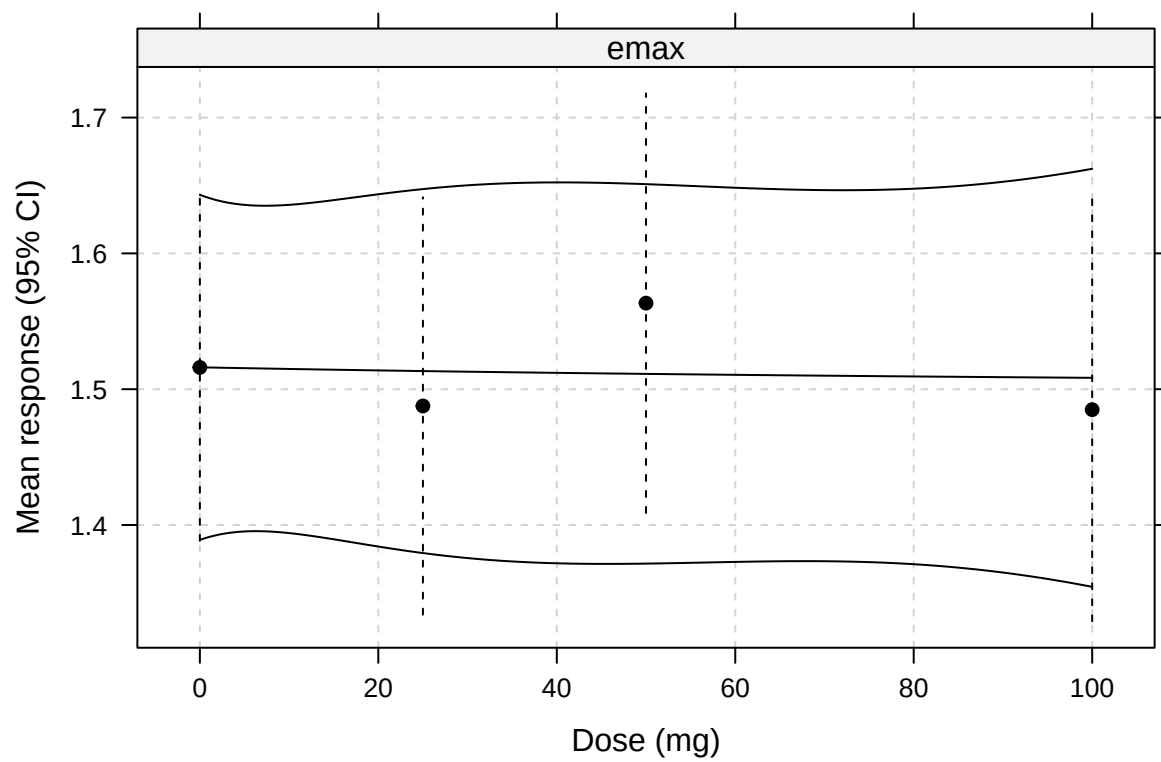
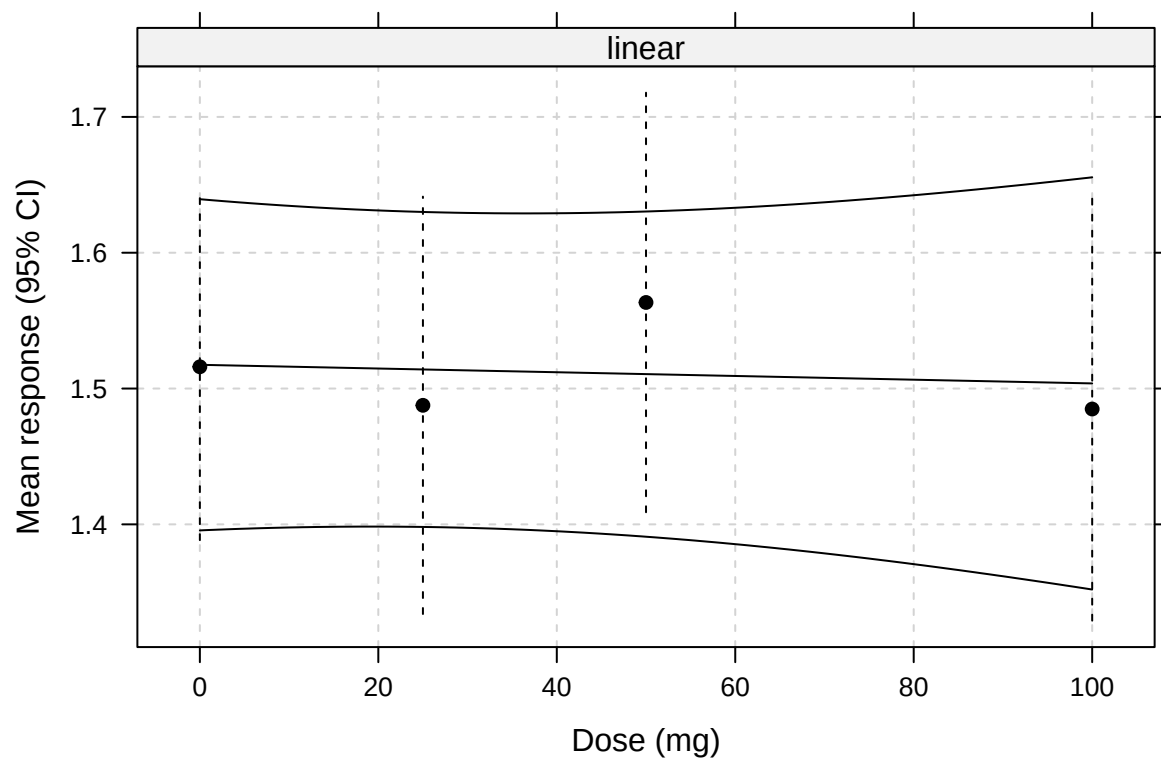
MCP-Mod tests

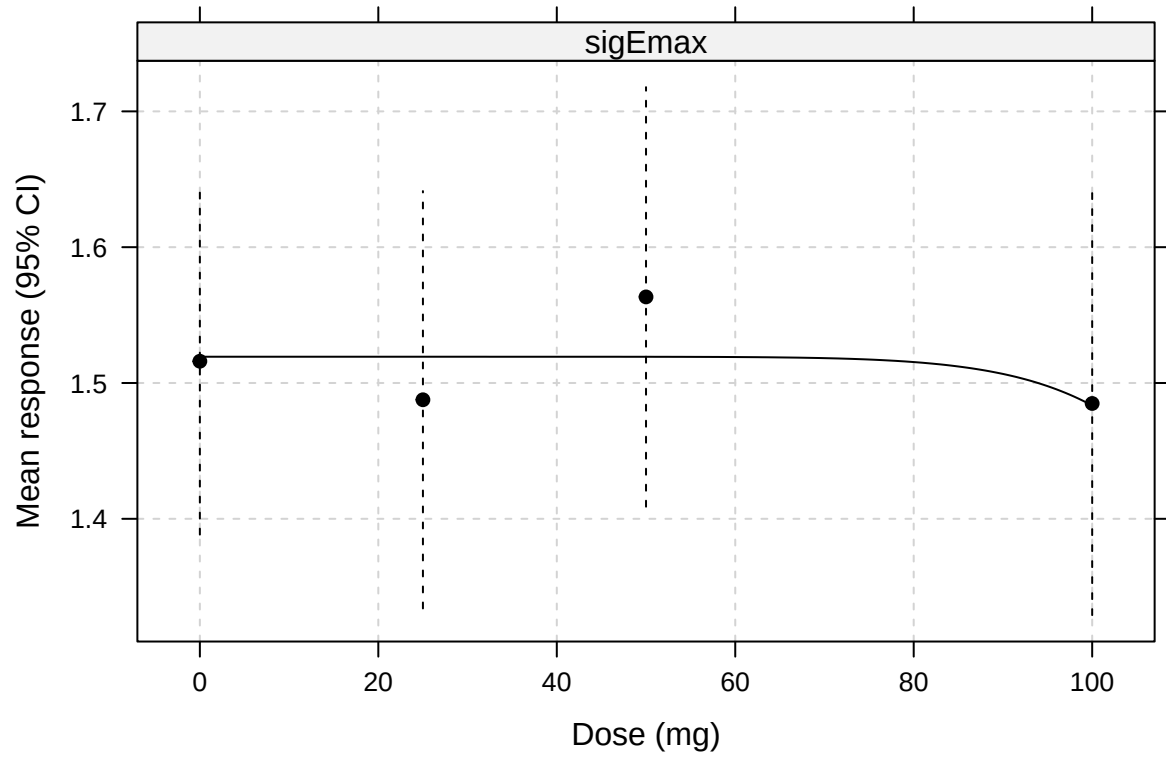
Candidate model	t-statistic	Adjusted p-value
sigEmax	0.20	0.58
emax	-0.03	0.67
linInt	-0.05	0.68
linear	-0.20	0.73

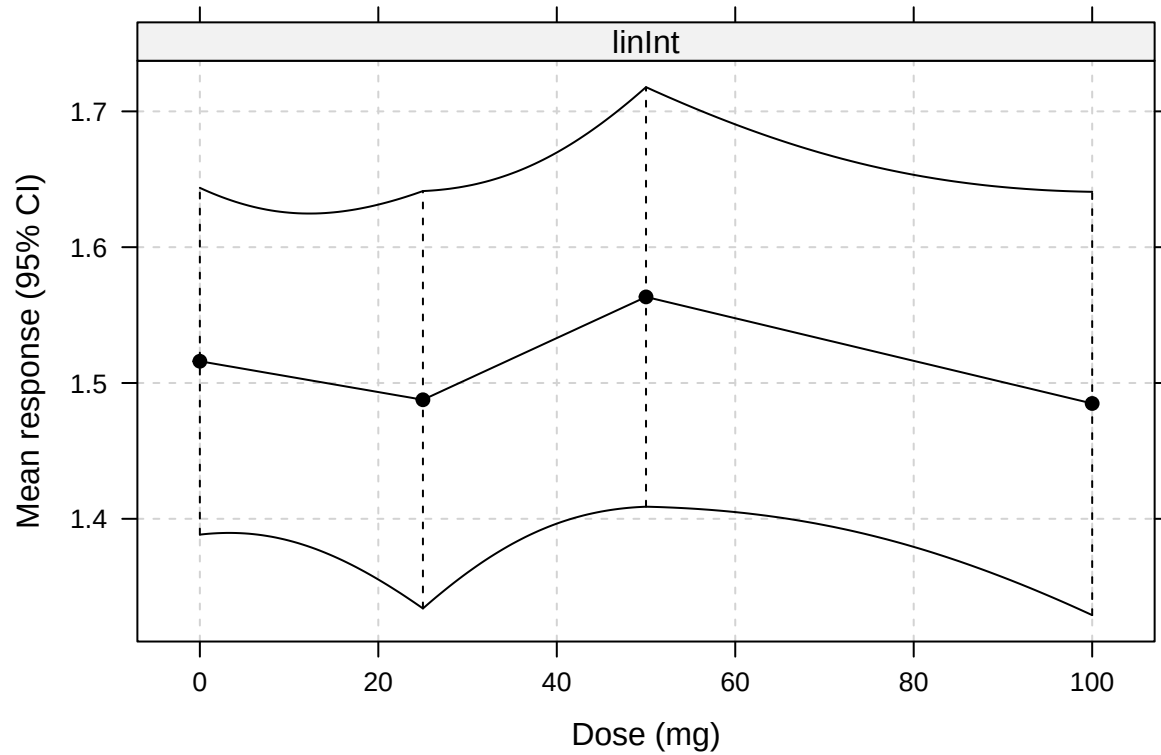
Fitted model parameters (AIC/ED)

Model	AIC-value	ED ₅₀	ED ₇₀	ED ₉₀
linear	5.36	50.00	70.00	90.00
emax	7.39	37.50	58.33	84.37
linInt	8.00	42.18	45.31	48.44
sigEmax	9.07	93.22	96.44	98.93

Fitted dose-response curves







Estimated means by dose

Treatment	Estimated mean (95% CI)
0 mg	1.52 (1.39 to 1.65)
25 mg	1.49 (1.33 to 1.64)
50 mg	1.56 (1.41 to 1.72)
100 mg	1.48 (1.33 to 1.64)

Mean relative cerebral blood flow (ROI2)

Optimal contrasts

Treatment	linear	emax	sigEmax	linInt
0 mg	-0.642	-0.815	-0.591	-0.866
25 mg	-0.206	0.023	-0.383	0.283
50 mg	0.120	0.294	0.366	0.294
100 mg	0.728	0.499	0.608	0.288

MCP-Mod tests

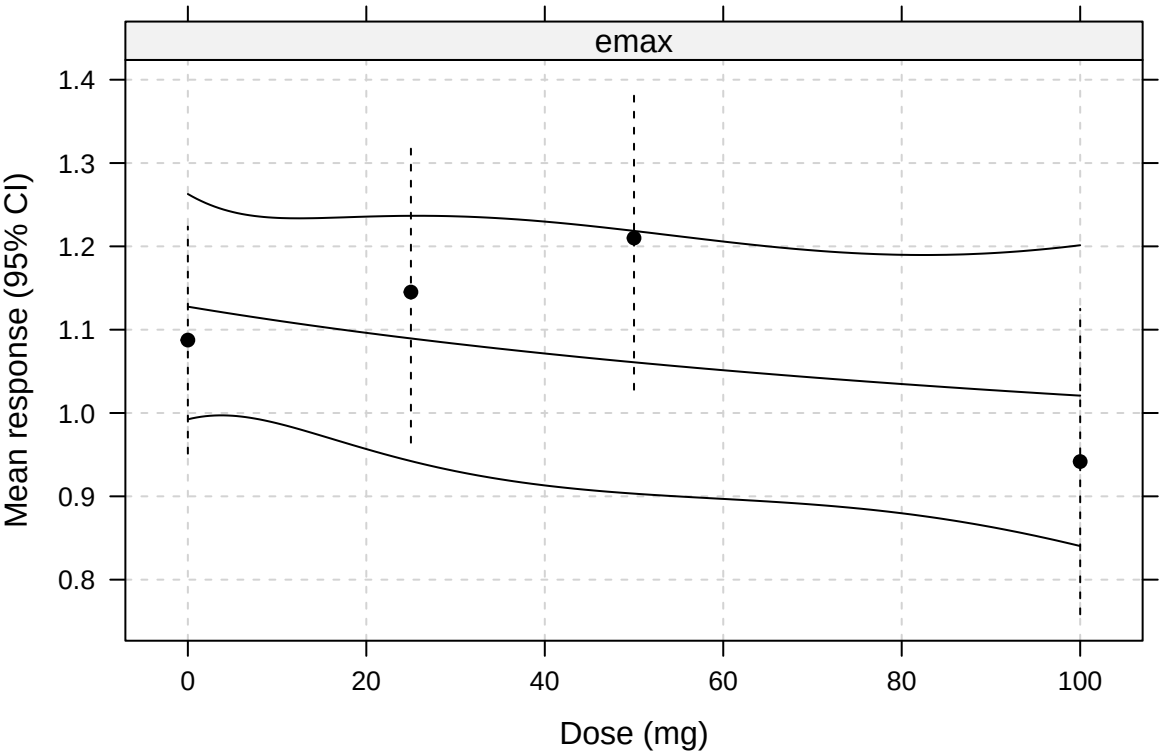
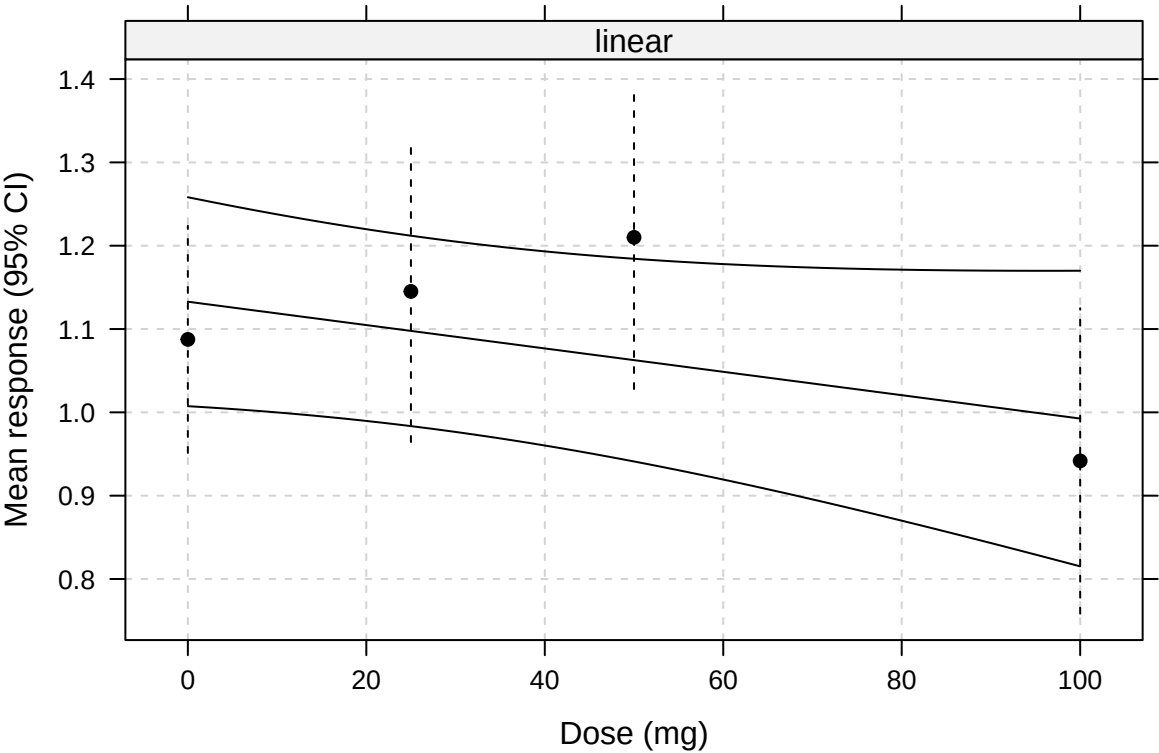
Candidate model	t-statistic	Adjusted p-value
linInt	0.15	0.6

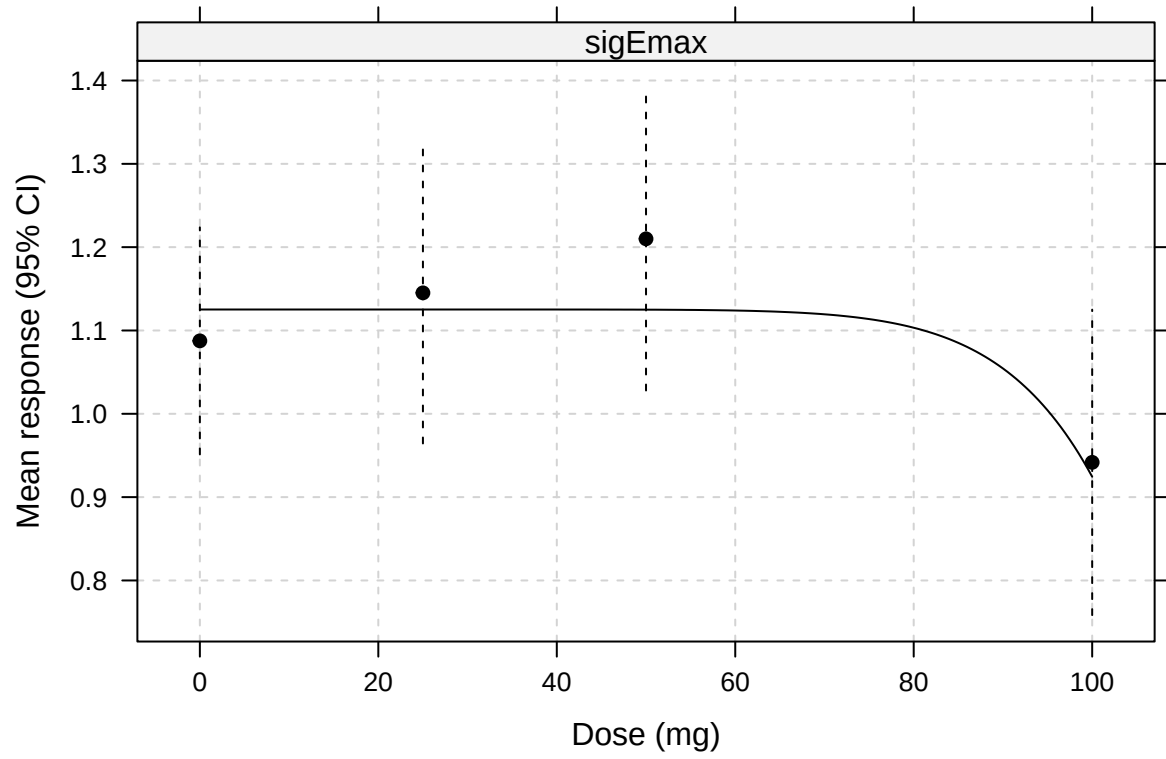
Candidate model	t-statistic	Adjusted p-value
emax	-0.52	0.83
sigEmax	-0.93	0.92
linear	-1.46	0.98

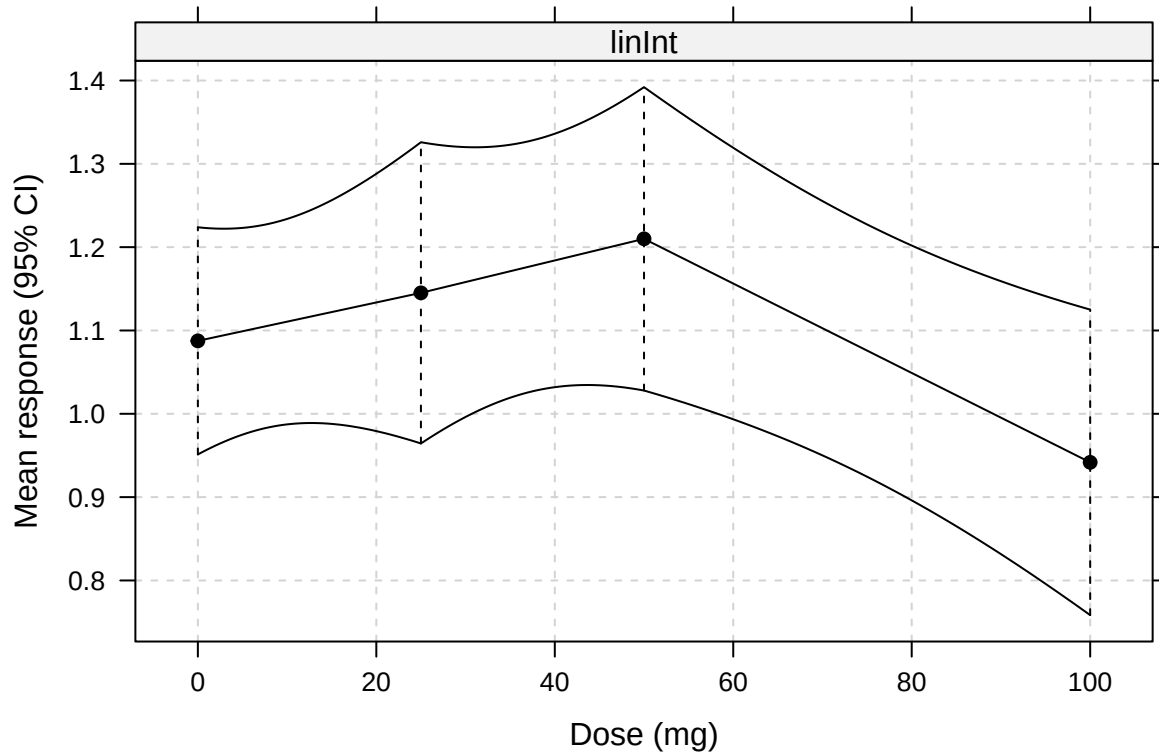
Fitted model parameters (AIC/ED)

Model	AIC-value	ED ₅₀	ED ₇₀	ED ₉₀
linInt	8.00	86.41	91.85	97.28
linear	9.04	50.00	70.00	90.00
sigEmax	9.56	93.22	96.45	98.93
emax	11.94	37.50	58.33	84.37

Fitted dose-response curves







Estimated means by dose

Treatment	Estimated mean (95% CI)
0 mg	1.09 (0.95 to 1.23)
25 mg	1.15 (0.96 to 1.33)
50 mg	1.21 (1.03 to 1.39)
100 mg	0.94 (0.76 to 1.13)

Mean relative solid stress (PTA-ROI)

Optimal contrasts

Treatment	linear	emax	sigEmax	linInt
0 mg	-0.629	-0.801	-0.587	-0.865
25 mg	-0.226	-0.014	-0.384	0.246
50 mg	0.121	0.292	0.349	0.303
100 mg	0.734	0.523	0.622	0.316

MCP-Mod tests

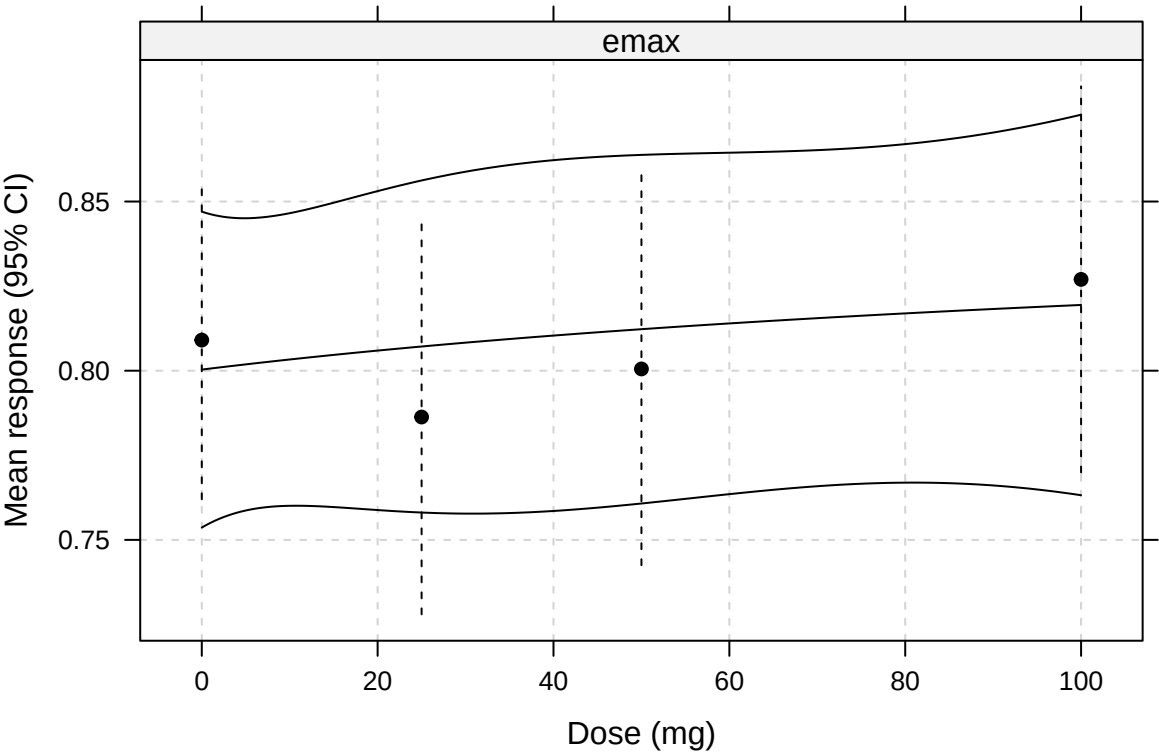
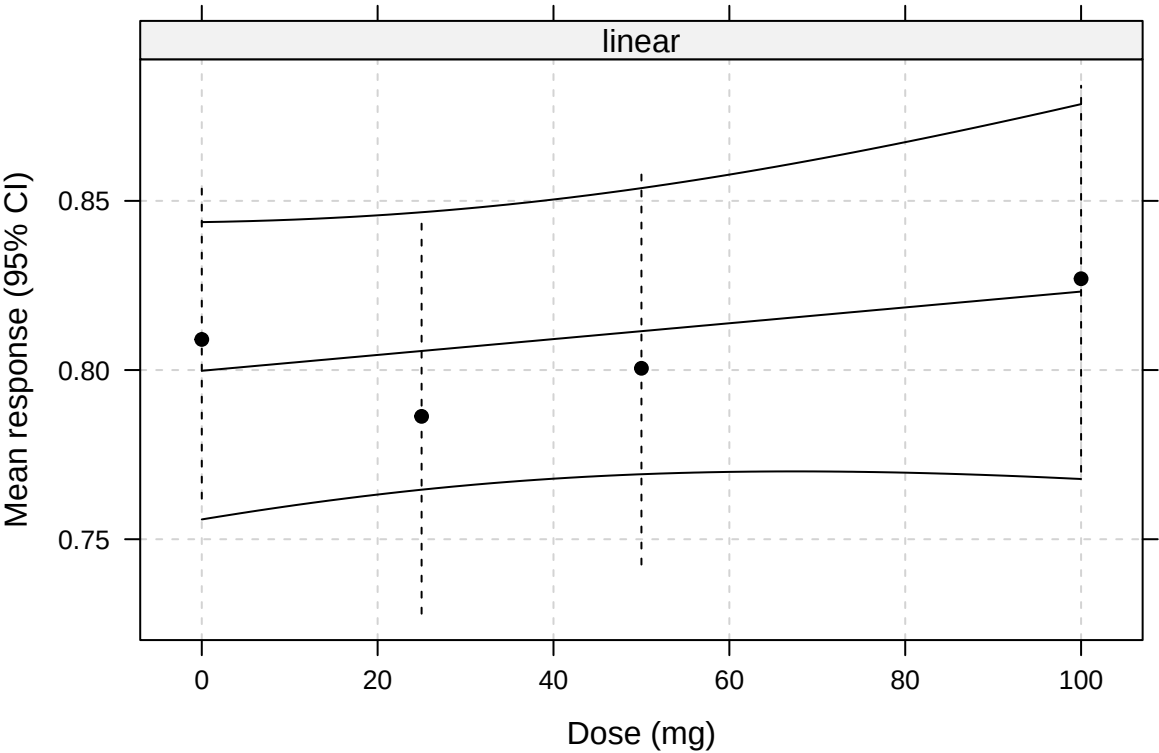
Candidate model	t-statistic	Adjusted p-value
linear	0.86	0.3

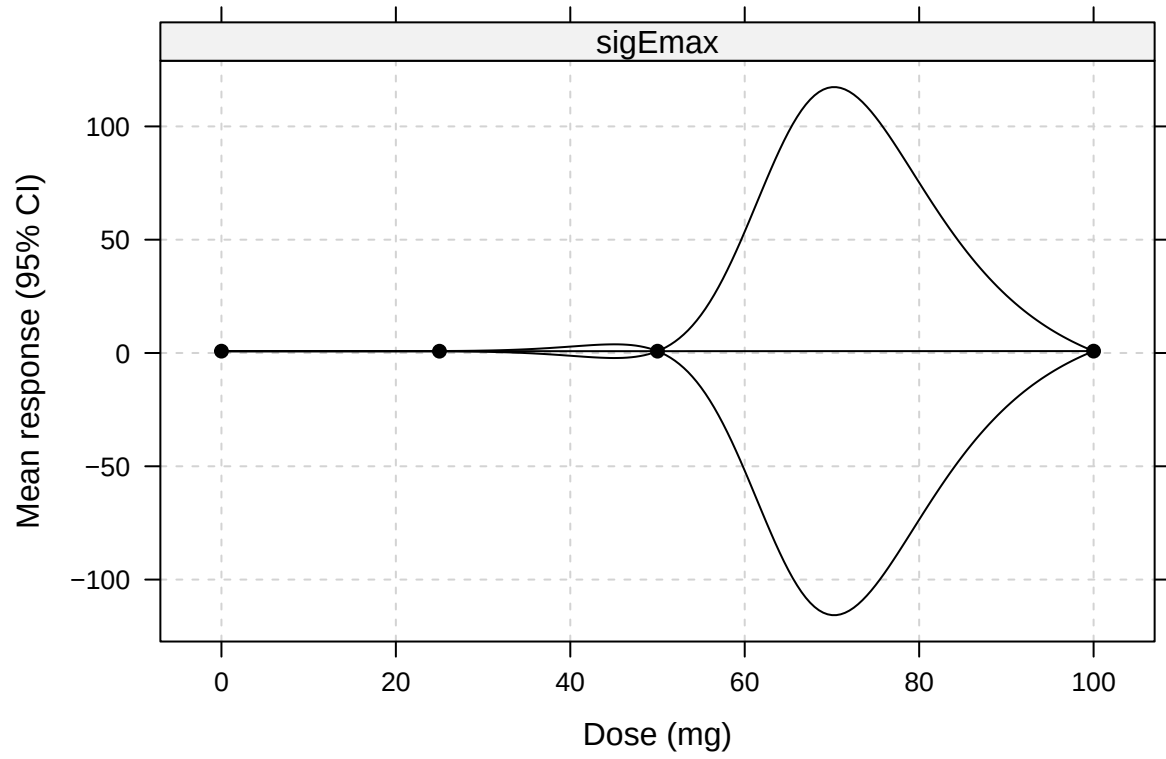
Candidate model	t-statistic	Adjusted p-value
sigEmax	0.84	0.31
emax	0.36	0.5
linInt	-0.13	0.7

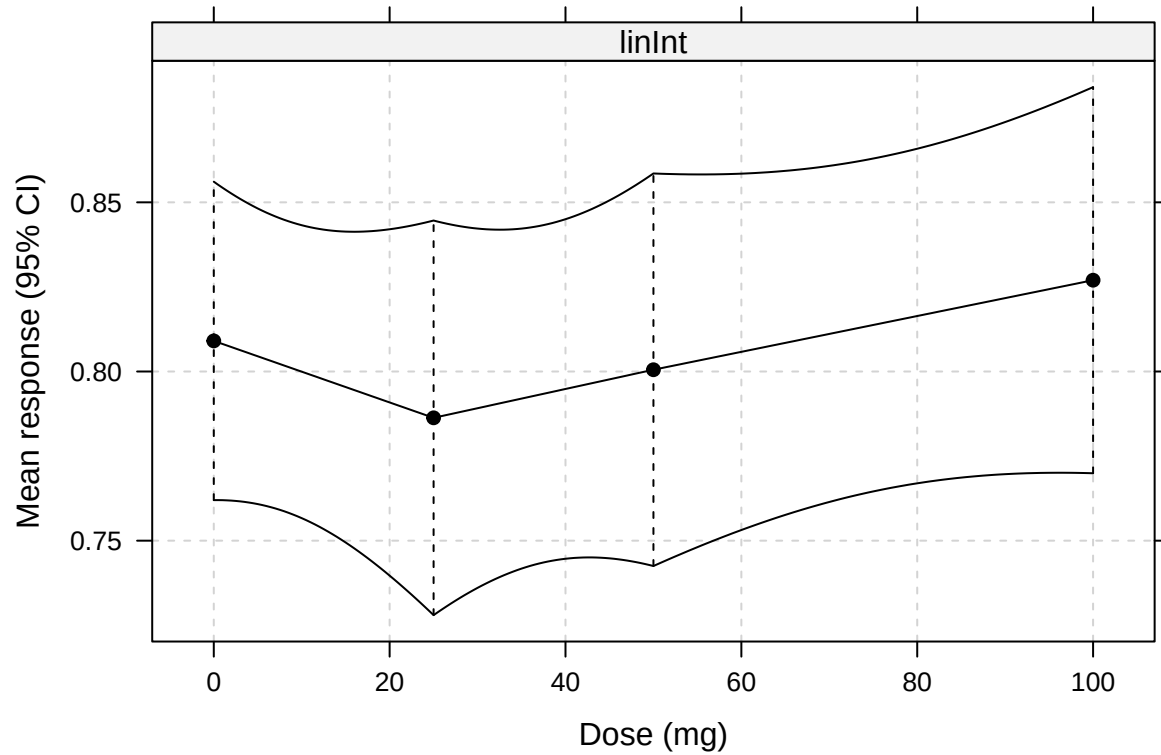
Fitted model parameters (AIC/ED)

Model	AIC-value	ED ₅₀	ED ₇₀	ED ₉₀
linear	5.16	50.00	70.00	90.00
emax	7.41	37.50	58.33	84.37
linInt	8.00	12.50	17.50	22.50
sigEmax	8.59	66.54	72.26	81.84

Fitted dose-response curves







Estimated means by dose

Treatment	Estimated mean (95% CI)
0 mg	0.81 (0.76 to 0.86)
25 mg	0.79 (0.73 to 0.85)
50 mg	0.8 (0.74 to 0.86)
100 mg	0.83 (0.77 to 0.88)

Mean relative solid stress (ROI2)

Optimal contrasts

Treatment	linear	emax	sigEmax	linInt
0 mg	-0.607	-0.794	-0.577	-0.864
25 mg	-0.236	-0.034	-0.397	0.229
50 mg	0.090	0.300	0.354	0.327
100 mg	0.753	0.528	0.620	0.307

MCP-Mod tests

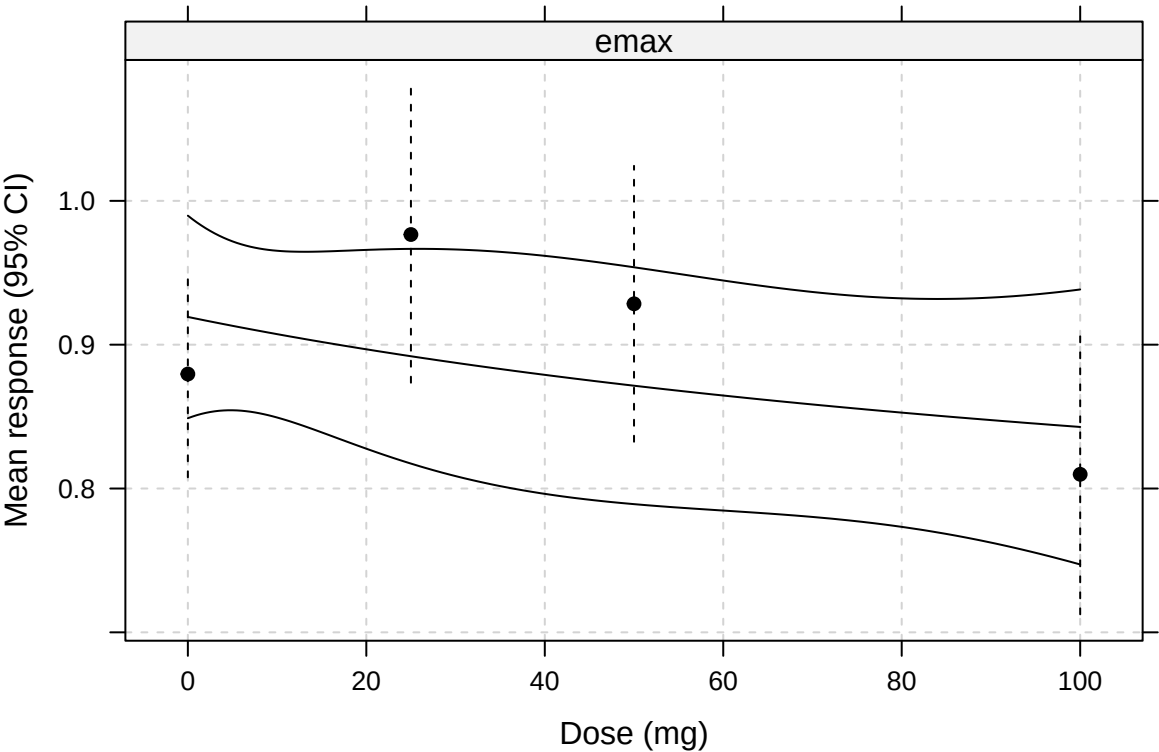
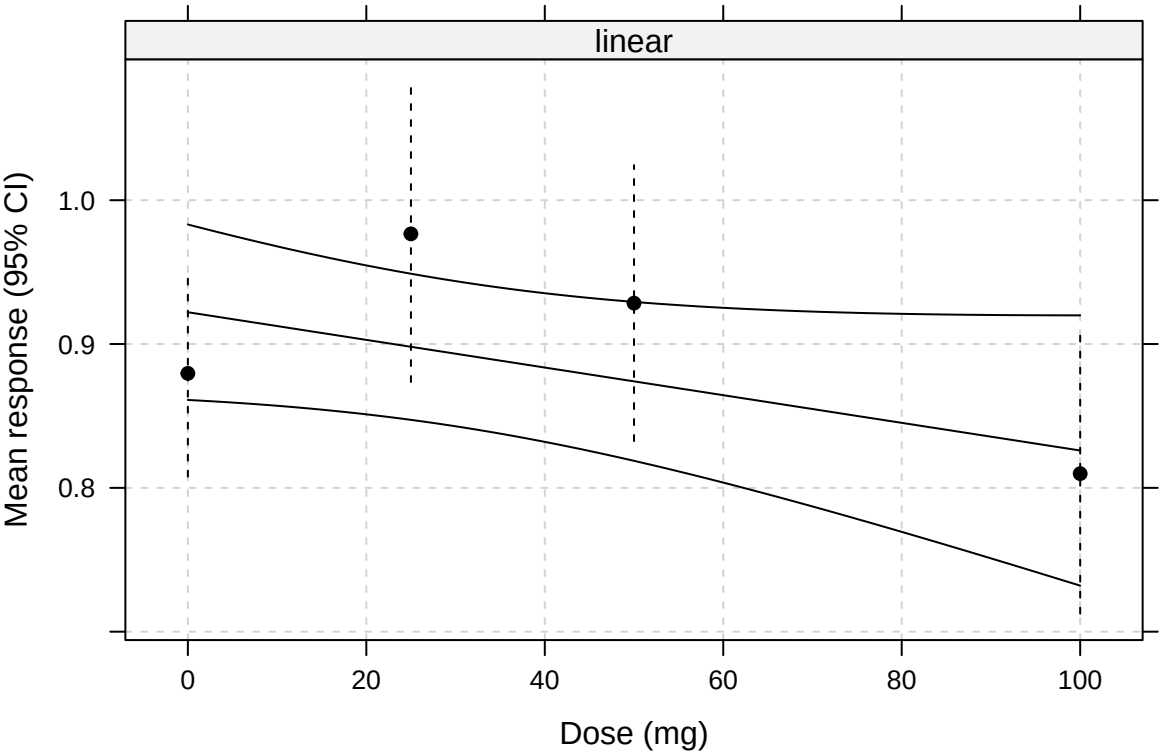
Candidate model	t-statistic	Adjusted p-value
linInt	0.39	0.5

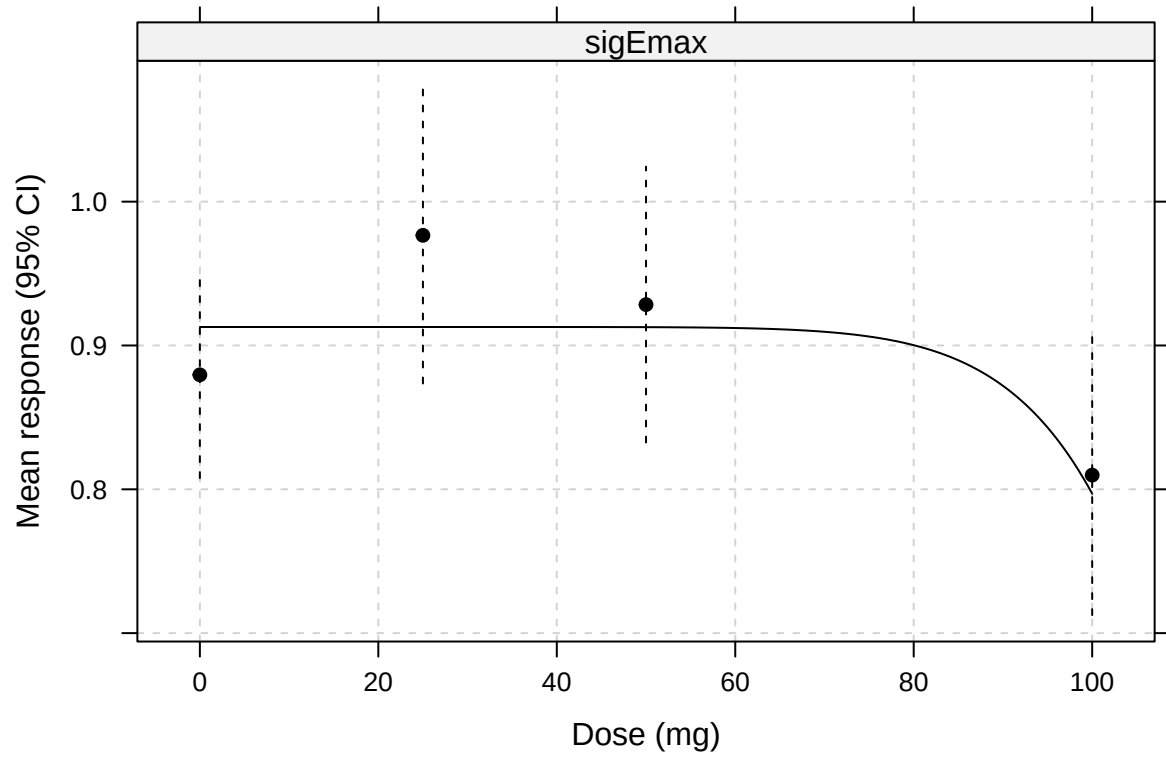
Candidate model	t-statistic	Adjusted p-value
emax	-0.60	0.85
sigEmax	-1.50	0.98
linear	-1.66	0.98

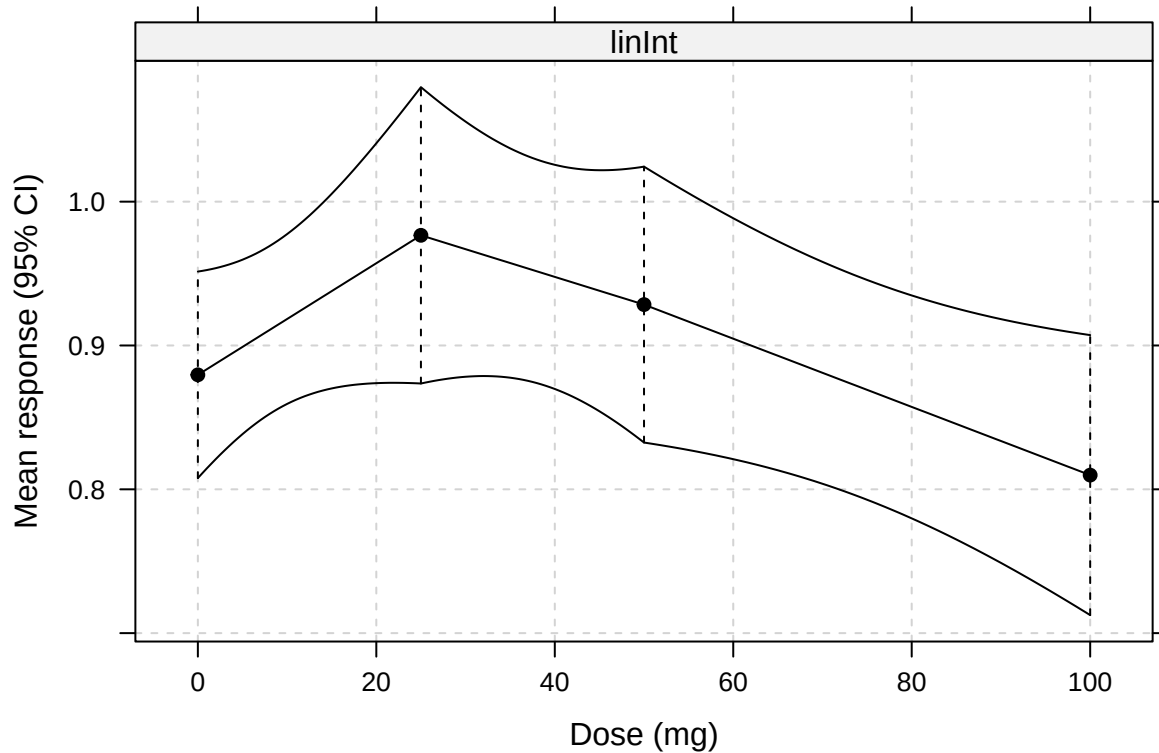
Fitted model parameters (AIC/ED)

Model	AIC-value	ED ₅₀	ED ₇₀	ED ₉₀
linInt	8.00	12.50	17.50	22.50
linear	8.93	50.00	70.00	90.00
sigEmax	10.26	93.22	96.44	98.93
emax	11.97	37.50	58.33	84.37

Fitted dose-response curves







Estimated means by dose

Treatment	Estimated mean (95% CI)
0 mg	0.88 (0.81 to 0.95)
25 mg	0.98 (0.87 to 1.08)
50 mg	0.93 (0.83 to 1.03)
100 mg	0.81 (0.71 to 0.91)

Other MRI Endpoints

For remaining MRI endpoints, we present estimated means by dose, contrasts vs 0 mg, and a plot with confidence intervals.

mrmeF

Means by dose

A tibble: 4 x 10

```

trted estimate std.error df conf.low conf.high statistic p.value trt
1 0 -0.0650 0.236 87.6 -0.533 0.403 -0.276 0.783 0 mg
2 25 0.0858 0.298 150. -0.503 0.674 0.288 0.774 25 mg
3 50 0.195 0.300 152. -0.397 0.788 0.652 0.516 50 mg
4 100 0.278 0.303 152. -0.320 0.875 0.917 0.361 100 mg
# i 1 more variable: mean_txt

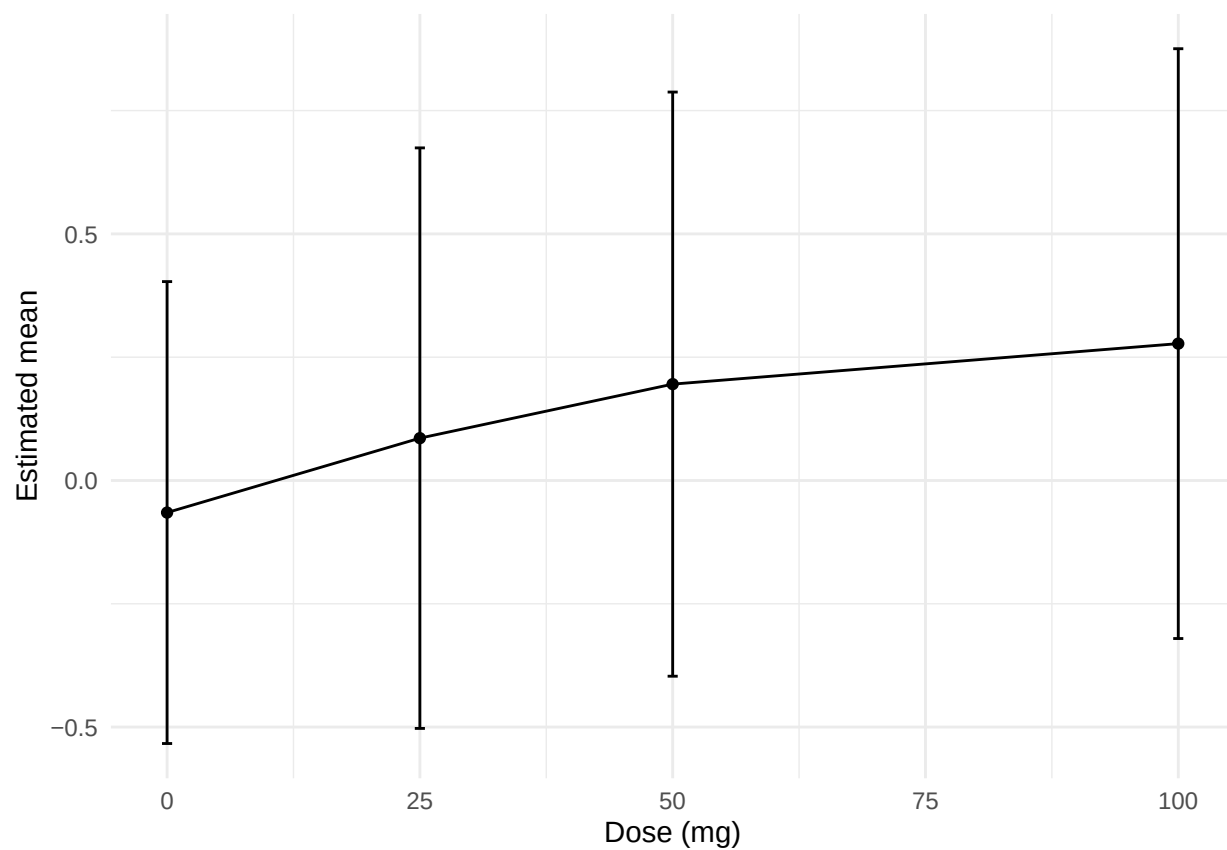
```

Differences vs 0 mg

A tibble: 3 x 11

```
term contrast null.value estimate std.error df conf.low conf.high 1 trtd 25 mg vs 0 mg 0 0.151 0.298 154.  
-0.559 0.860 2 trtd 50 mg vs 0 mg 0 0.260 0.296 154. -0.445 0.966 3 trtd 100 mg vs 0 mg 0 0.343 0.300 157.  
-0.373 1.06 # i 3 more variables: statistic , adj.p.value , estimate_txt
```

Plot



mrdsc

Means by dose

A tibble: 4 x 10

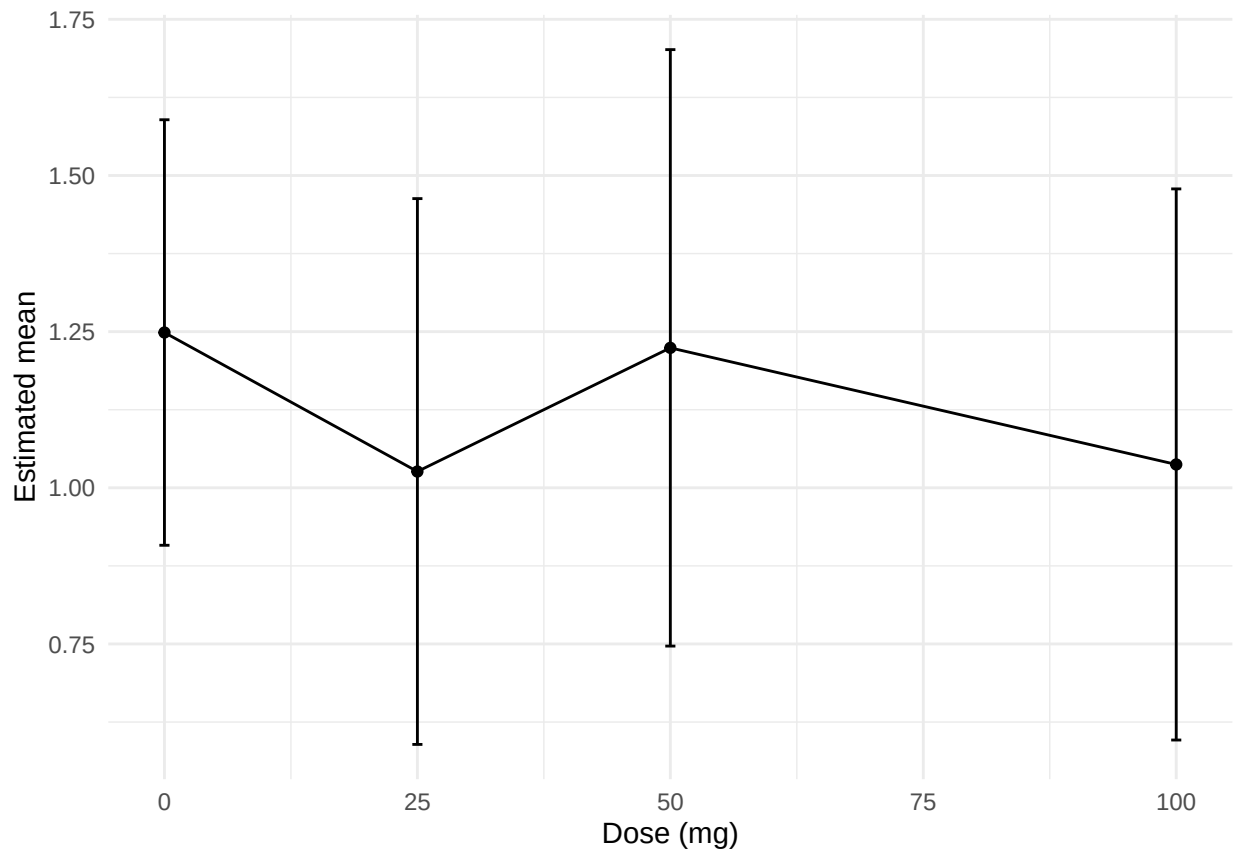
```
trtcd estimate std.error df conf.low conf.high statistic p.value trt
1 0 1.25 0.170 51.7 0.908 1.59 7.36 1.37e-9 0 mg 2 25 1.03 0.220 80.2 0.589 1.46 4.67 1.18e-5 25 mg 3 50 1.22
0.240 86.8 0.747 1.70 5.10 2.00e-6 50 mg 4 100 1.04 0.222 81.5 0.596 1.48 4.68 1.13e-5 100 ~ # i 1 more
variable: mean_txt
```

Differences vs 0 mg

A tibble: 3 x 11

```
term contrast null.value estimate std.error df conf.low conf.high 1 trtcd 25 mg vs 0 mg 0 -0.223 0.239 77.3
-0.799 0.354 2 trtcd 50 mg vs 0 mg 0 -0.0245 0.252 76.3 -0.632 0.583 3 trtcd 100 mg vs 0 mg 0 -0.211 0.235
75.3 -0.778 0.355 # i 3 more variables: statistic , adj.p.value , estimate_txt
```

Plot



mrdpta

Means by dose

A tibble: 4 x 10

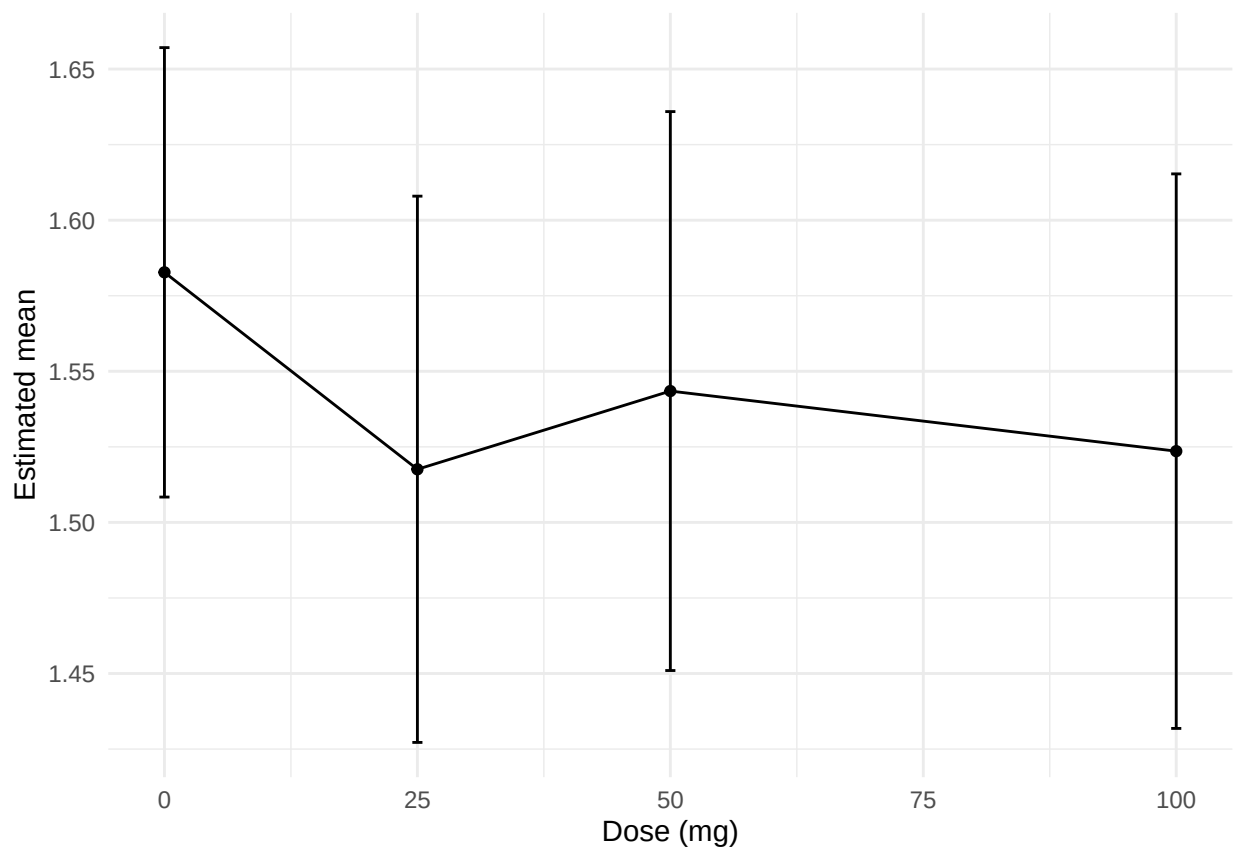
```
trtcd estimate std.error df conf.low conf.high statistic p.value trt
1 0 1.58 0.0374 82.2 1.51 1.66 42.3 1.30e-57 0 mg
2 25 1.52 0.0457 138. 1.43 1.61 33.2 8.83e-68 25 mg 3 50 1.54 0.0468 143. 1.45 1.64 33.0 1.28e-68 50 mg 4 100
1.52 0.0464 141. 1.43 1.62 32.8 5.45e-68 100 mg # i 1 more variable: mean_txt
```

Differences vs 0 mg

A tibble: 3 x 11

```
term contrast null.value estimate std.error df conf.low conf.high 1 trtcd 25 mg vs 0 mg 0 -0.0652 0.0429 153.
-0.167 0.0371 2 trtcd 50 mg vs 0 mg 0 -0.0393 0.0443 156. -0.145 0.0663 3 trtcd 100 mg vs 0 mg 0 -0.0592
0.0438 155. -0.164 0.0452 # i 3 more variables: statistic , adj.p.value , estimate_txt
```

Plot



mrpta

Means by dose

A tibble: 4 x 10

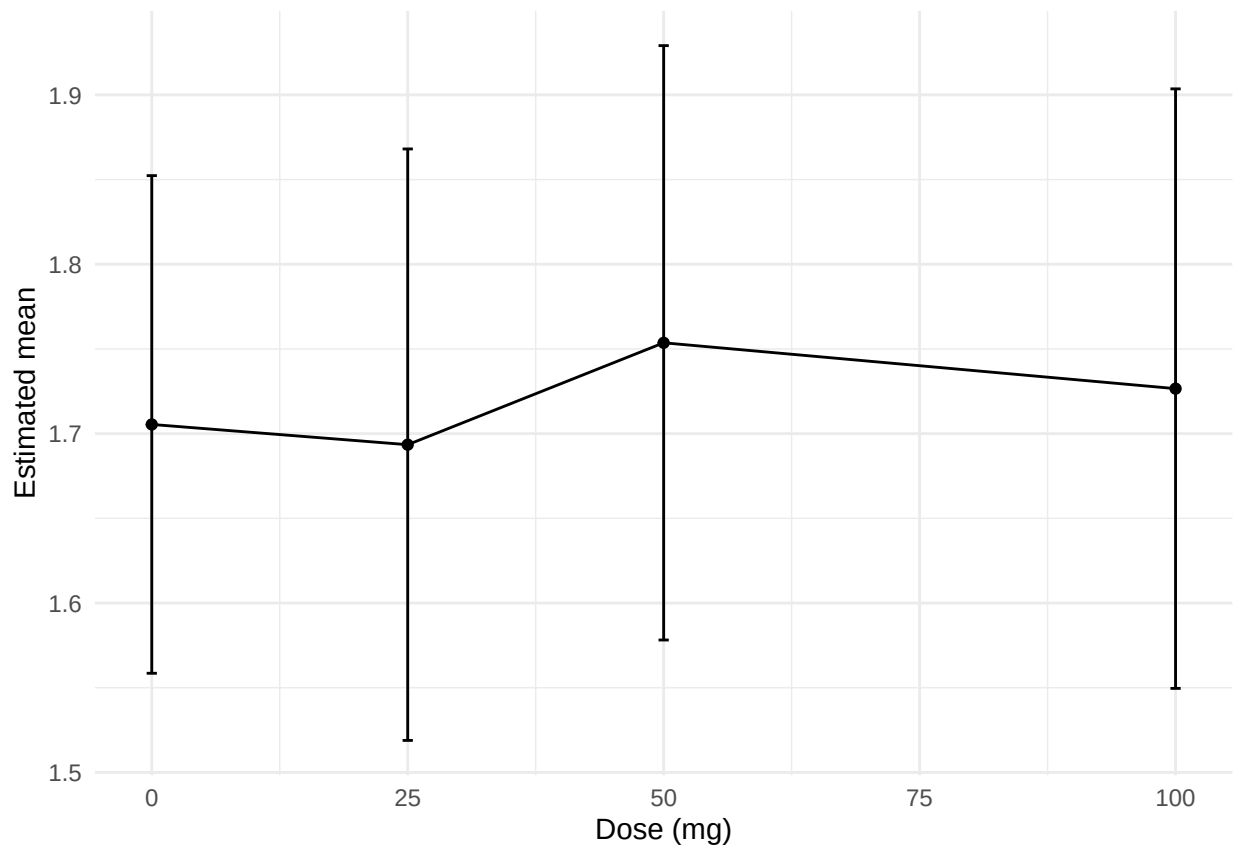
```
trtcd estimate std.error df conf.low conf.high statistic p.value trt
1 0 1.71 0.0738 76.7 1.56 1.85 23.1 2.55e-36 0 mg
2 25 1.69 0.0882 128. 1.52 1.87 19.2 1.97e-39 25 mg 3 50 1.75 0.0887 129. 1.58 1.93 19.8 6.70e-41 50 mg 4 100
1.73 0.0895 131. 1.55 1.90 19.3 4.51e-40 100 mg # i 1 more variable: mean_txt
```

Differences vs 0 mg

A tibble: 3 x 11

```
term contrast null.value estimate std.error df conf.low conf.high 1 trtcd 25 mg vs 0 mg 0 -0.0119 0.0787 148.
-0.200 0.176 2 trtcd 50 mg vs 0 mg 0 0.0482 0.0782 148. -0.138 0.235 3 trtcd 100 mg vs 0 mg 0 0.0211 0.0796
150. -0.169 0.211 # i 3 more variables: statistic , adj.p.value , estimate_txt
```

Plot



mrrdv

Means by dose

A tibble: 4 x 10

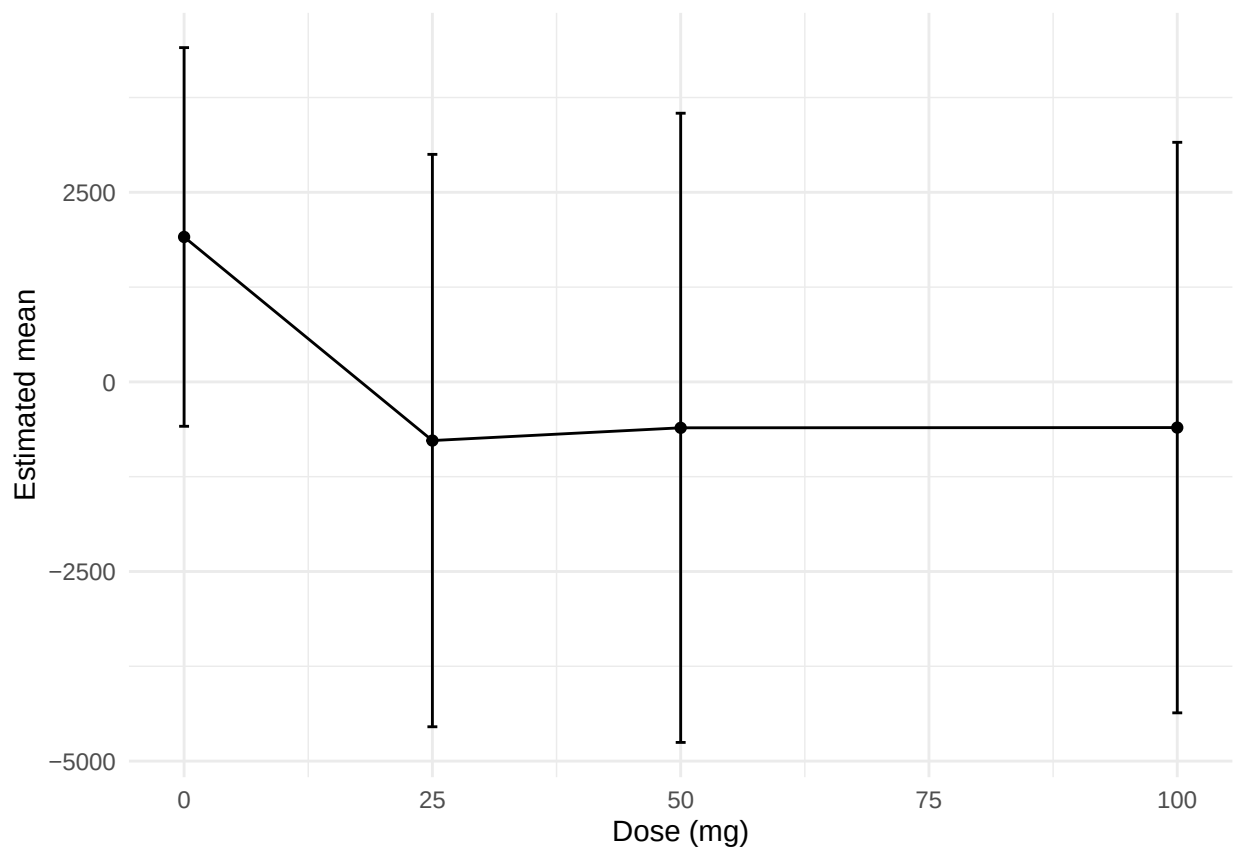
```
trtcd estimate std.error df conf.low conf.high statistic p.value trt
1 0 1911. 1247. 57.3 -585. 4408. 1.53 0.131 0 mg
2 25 -774. 1895. 75.7 -4548. 3000. -0.408 0.684 25 mg
3 50 -605. 2083. 76.7 -4754. 3543. -0.291 0.772 50 mg
4 100 -602. 1879. 58.5 -4364. 3159. -0.320 0.750 100 mg # i 1 more variable: mean_txt
```

Differences vs 0 mg

A tibble: 3 x 11

```
term contrast null.value estimate std.error df conf.low conf.high 1 trtcd 25 mg vs 0 mg 0 -2685. 2558. 84.6
-8833. 3463. 2 trtcd 50 mg vs 0 mg 0 -2517. 2671. 86.4 -8935. 3902. 3 trtcd 100 mg vs 0 mg 0 -2514. 2522.
76.1 -8589. 3562. # i 3 more variables: statistic , adj.p.value , estimate_txt
```

Plot



mrdrfa

Means by dose

A tibble: 4 x 10

```
trtcd estimate std.error df conf.low conf.high statistic p.value trt
1 0 0.466 0.0159 97.4 0.434 0.497 29.4 3.52e-50 0 mg
2 25 0.480 0.0209 163. 0.439 0.521 22.9 7.89e-53 25 mg
3 50 0.446 0.0214 163. 0.403 0.488 20.8 8.21e-48 50 mg
4 100 0.481 0.0213 164. 0.439 0.523 22.6 3.29e-52 100 mg
```

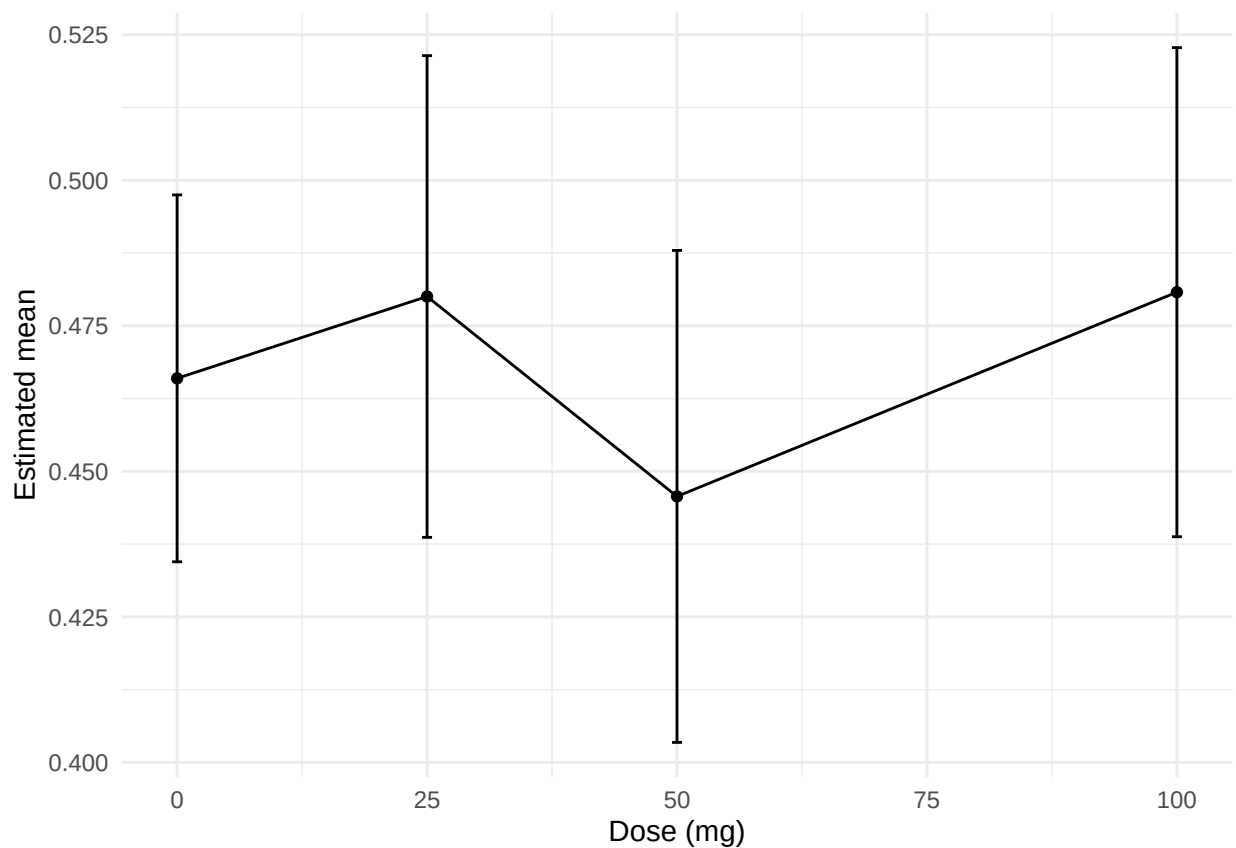
i 1 more variable: mean_txt

Differences vs 0 mg

A tibble: 3 x 11

```
term contrast null.value estimate std.error df conf.low conf.high 1 trtcd 25 mg vs 0 mg 0 0.0141 0.0222 160.
-0.0388 0.0669 2 trtcd 50 mg vs 0 mg 0 -0.0203 0.0227 165. -0.0744 0.0338 3 trtcd 100 mg vs 0 mg 0 0.0148
0.0225 163. -0.0388 0.0684 # i 3 more variables: statistic , adj.p.value , estimate_txt
```

Plot



mrdrsi

Means by dose

A tibble: 4 x 10

```
trtcd estimate std.error df conf.low conf.high statistic p.value trt
1 0 0.455 0.0223 90.6 0.411 0.499 20.4 1.19e-35 0 mg
2 25 0.499 0.0289 154. 0.442 0.556 17.2 9.38e-38 25 mg
3 50 0.445 0.0297 156. 0.386 0.504 15.0 5.01e-32 50 mg
4 100 0.493 0.0295 157. 0.435 0.552 16.7 1.03e-36 100 mg
```

 # i 1 more variable: mean_txt

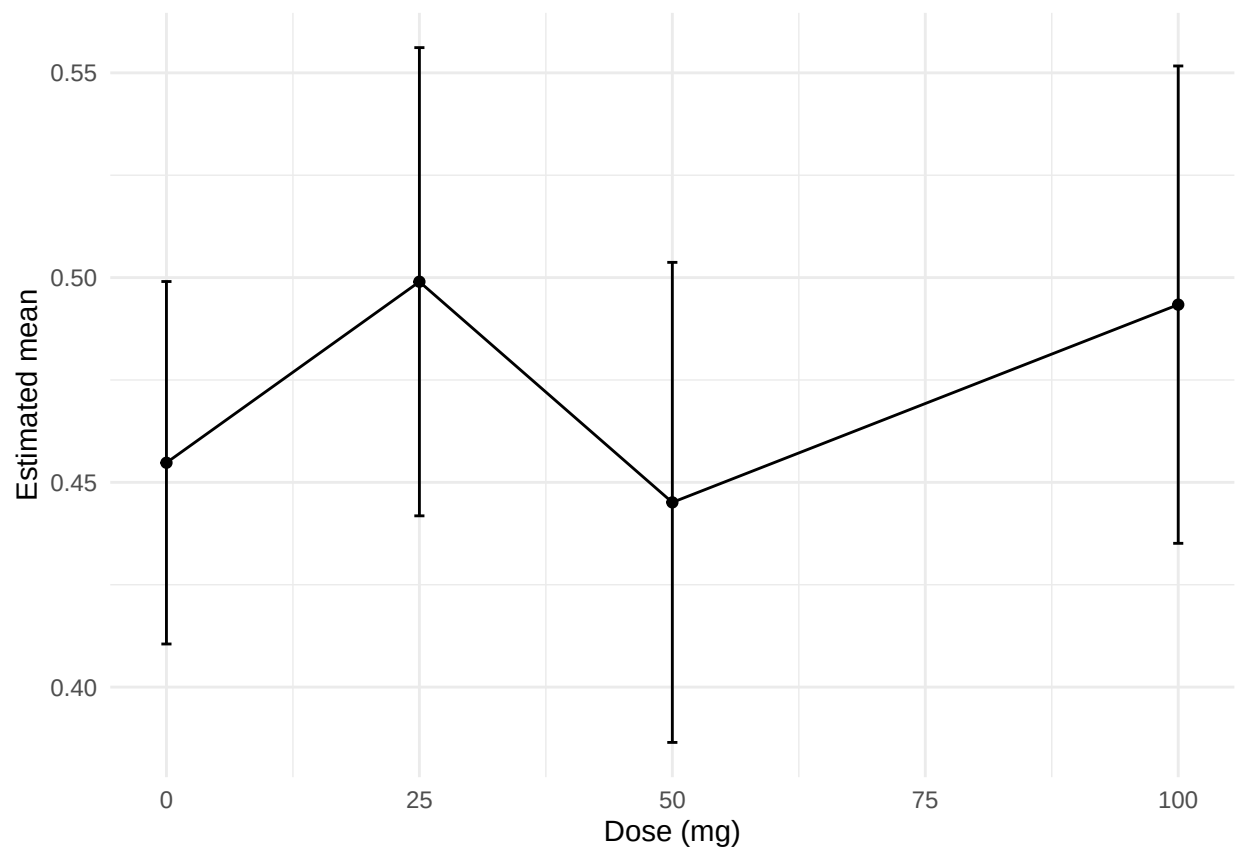
Differences vs 0 mg

A tibble: 3 x 11

```
term contrast null.value estimate std.error df conf.low conf.high 1 trtcd 25 mg vs 0 mg 0 0.0442 0.0295 154.
-0.0261 0.115 2 trtcd 50 mg vs 0 mg 0 -0.00968 0.0304 158. -0.0821 0.0627 3 trtcd 100 mg vs 0 mg 0 0.0386
0.0300 154. -0.0329 0.110
```

 # i 3 more variables: statistic , adj.p.value , estimate_txt

Plot



mrtrt

Means by dose

A tibble: 4 x 10

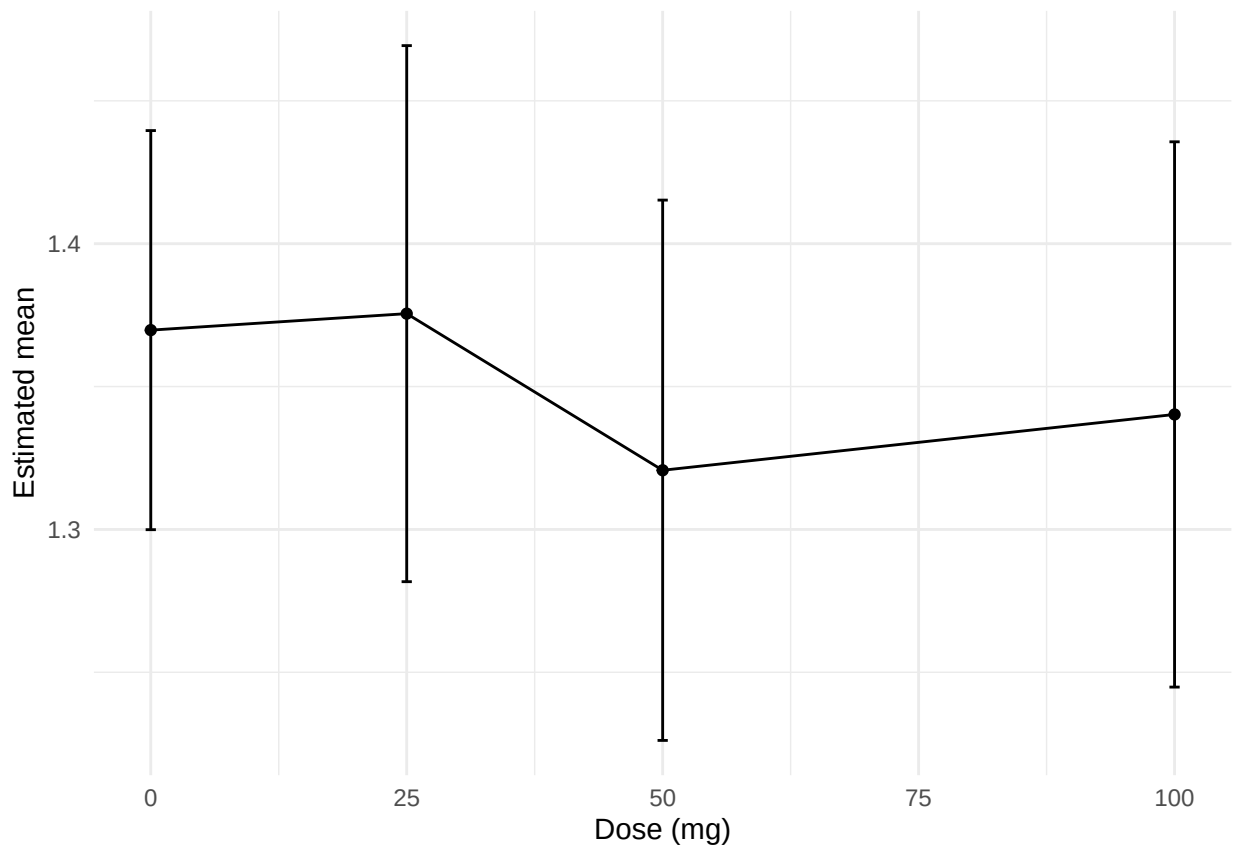
```
trtcd estimate std.error df conf.low conf.high statistic p.value trt
1 0 1.37 0.0352 104. 1.30 1.44 38.9 8.85e-64 0 mg
2 25 1.38 0.0475 169. 1.28 1.47 28.9 2.55e-67 25 mg 3 50 1.32 0.0479 170. 1.23 1.42 27.6 1.21e-64 50 mg 4 100
1.34 0.0483 168. 1.24 1.44 27.7 1.25e-64 100 mg # i 1 more variable: mean_txt
```

Differences vs 0 mg

A tibble: 3 x 11

```
term contrast null.value estimate std.error df conf.low conf.high 1 trtcd 25 mg vs 0 mg 0 0.00577 0.0525 165.
-0.119 0.131 2 trtcd 50 mg vs 0 mg 0 -0.0490 0.0522 164. -0.173 0.0754 3 trtcd 100 mg vs 0 mg 0 -0.0295
0.0529 168. -0.155 0.0964 # i 3 more variables: statistic , adj.p.value , estimate_txt
```

Plot



mrrvc

Means by dose

A tibble: 4 x 10

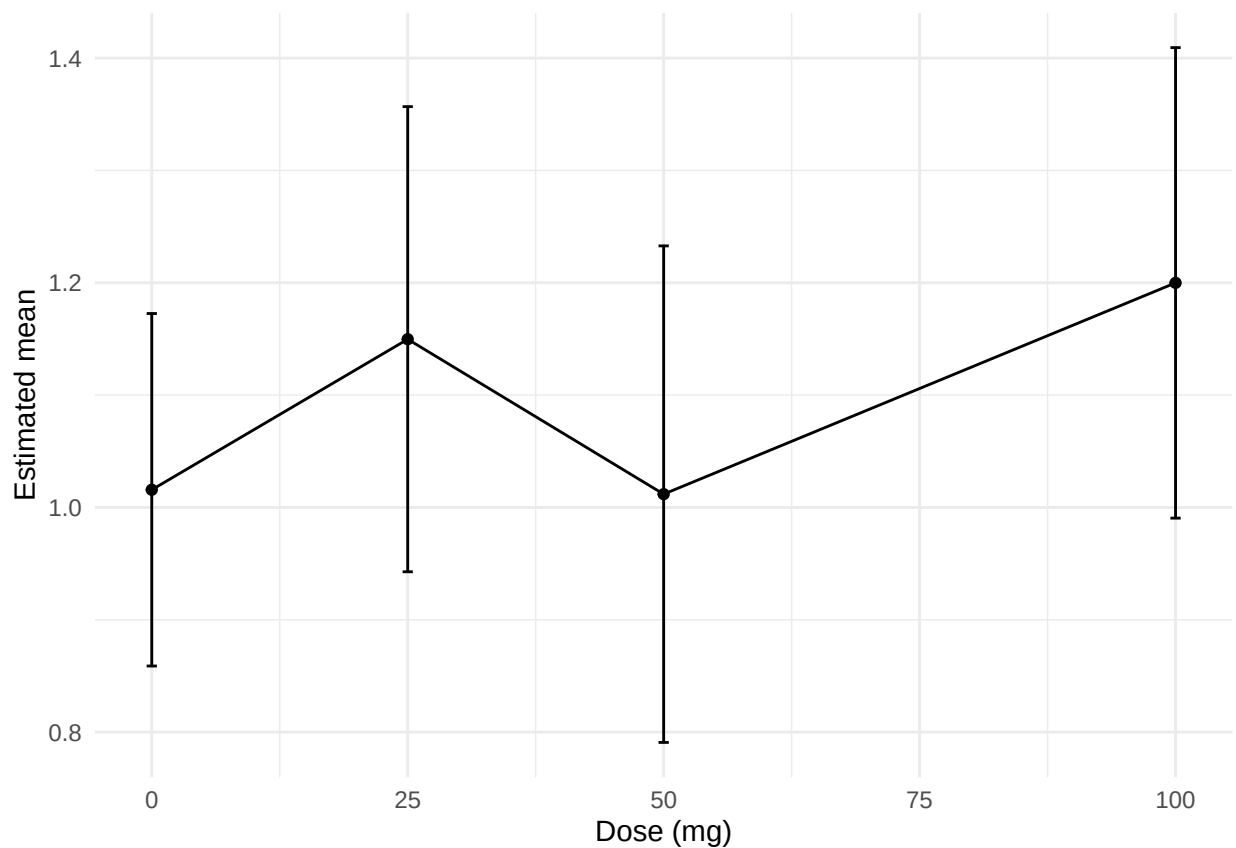
```
trtcd estimate std.error df conf.low conf.high statistic p.value trt
1 0 1.02 0.0784 57.9 0.859 1.17 13.0 9.27e-19 0 mg
2 25 1.15 0.104 84.7 0.943 1.36 11.0 4.43e-18 25 mg 3 50 1.01 0.111 88.0 0.791 1.23 9.10 2.56e-14 50 mg 4 100
1.20 0.105 85.8 0.990 1.41 11.4 7.54e-19 100 mg # i 1 more variable: mean_txt
```

Differences vs 0 mg

A tibble: 3 x 11

```
term contrast null.value estimate std.error df conf.low conf.high 1 trtcd 25 mg vs 0 mg 0 0.134 0.119 80.8
-0.152 0.420 2 trtcd 50 mg vs 0 mg 0 -0.00390 0.123 81.9 -0.300 0.293 3 trtcd 100 mg vs 0 mg 0 0.184 0.117
78.8 -0.0972 0.466 # i 3 more variables: statistic , adj.p.value , estimate_txt
```

Plot



mrrvsi

Means by dose

A tibble: 4 x 10

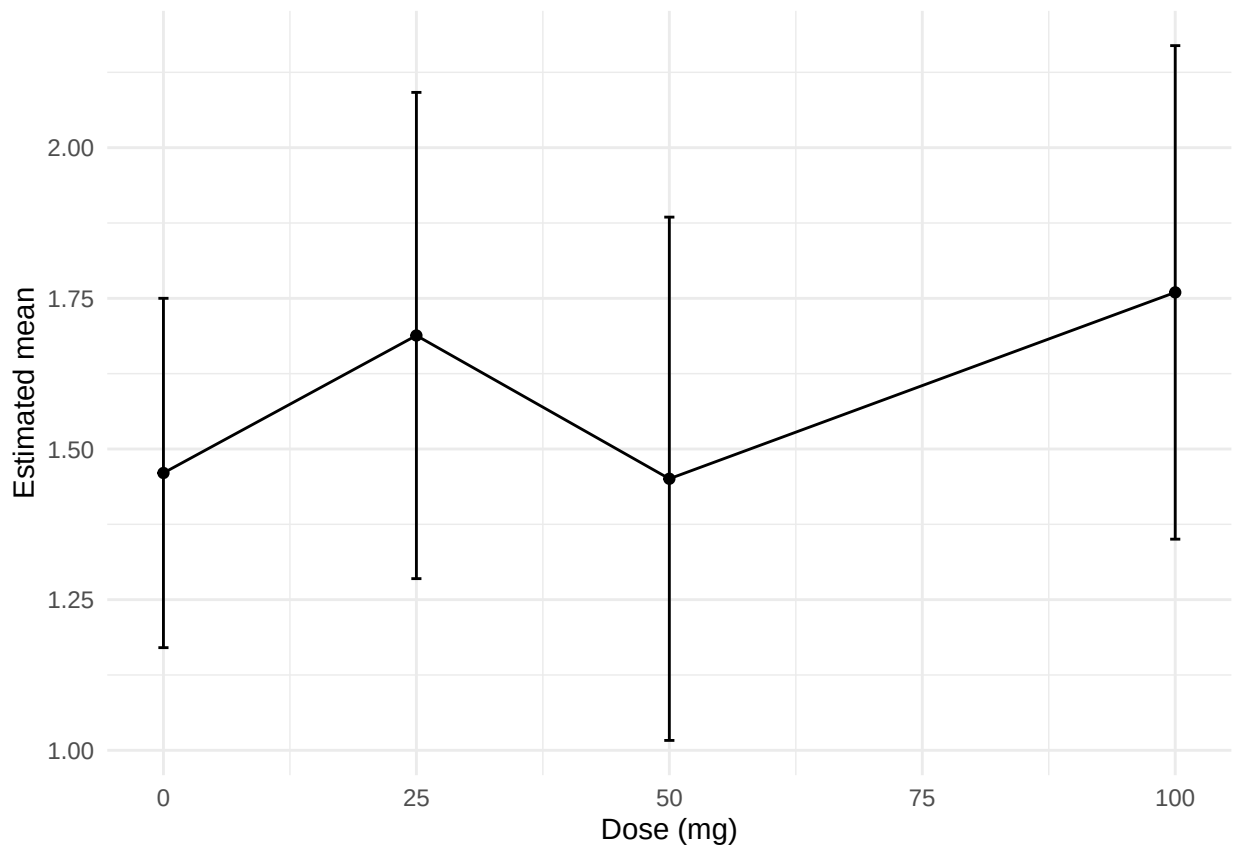
```
trtcd estimate std.error df conf.low conf.high statistic p.value trt
1 0 1.46 0.145 64.4 1.17 1.75 10.1 7.73e-15 0 mg
2 25 1.69 0.203 86.2 1.28 2.09 8.32 1.17e-12 25 mg 3 50 1.45 0.219 89.0 1.02 1.88 6.64 2.42e- 9 50 mg 4 100
1.76 0.206 86.4 1.35 2.17 8.54 4.06e-13 100 mg # i 1 more variable: mean_txt
```

Differences vs 0 mg

A tibble: 3 x 11

```
term contrast null.value estimate std.error df conf.low conf.high 1 trtcd 25 mg vs 0 mg 0 0.228 0.249 88.4
-0.369 0.825 2 trtcd 50 mg vs 0 mg 0 -0.00958 0.257 88.6 -0.628 0.609 3 trtcd 100 mg vs 0 mg 0 0.300 0.246
87.2 -0.291 0.891 # i 3 more variables: statistic , adj.p.value , estimate_txt
```

Plot



mrrvt

Means by dose

A tibble: 4 x 10

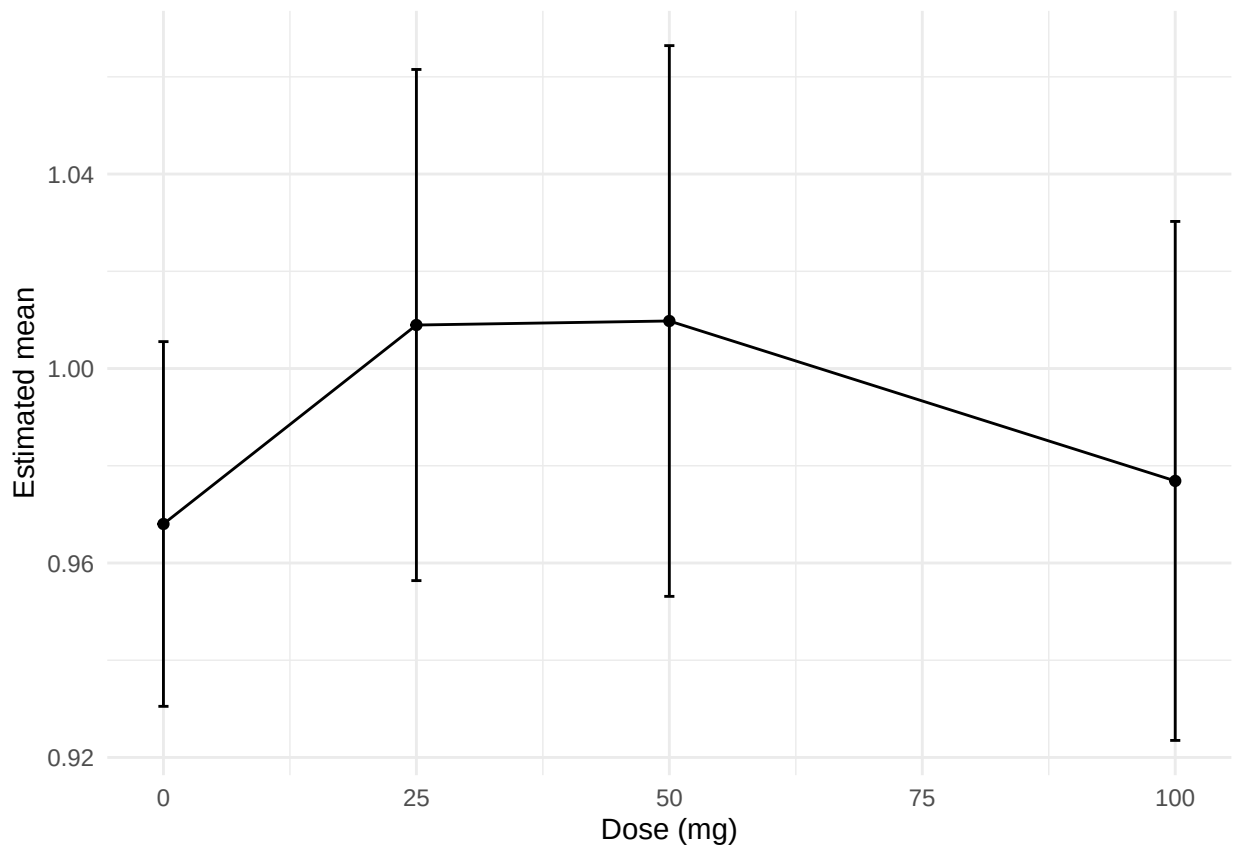
```
trtcd estimate std.error df conf.low conf.high statistic p.value trt
1 0 0.968 0.0188 65.0 0.931 1.01 51.5 1.94e-54 0 mg
2 25 1.01 0.0264 86.1 0.956 1.06 38.2 9.81e-56 25 mg 3 50 1.01 0.0285 88.9 0.953 1.07 35.4 3.27e-54 50 mg 4
100 0.977 0.0268 86.0 0.924 1.03 36.4 5.29e-54 100 mg # i 1 more variable: mean_txt
```

Differences vs 0 mg

A tibble: 3 x 11

```
term contrast null.value estimate std.error df conf.low conf.high 1 trtcd 25 mg vs 0 mg 0 0.0409 0.0327 89.4
-0.0375 0.119 2 trtcd 50 mg vs 0 mg 0 0.0418 0.0338 89.5 -0.0395 0.123 3 trtcd 100 mg vs 0 mg 0 0.00886
0.0324 88.4 -0.0689 0.0866 # i 3 more variables: statistic , adj.p.value , estimate_txt
```

Plot



mrkte

Means by dose

A tibble: 4 x 10

```
trtcd estimate std.error df conf.low conf.high statistic p.value trt
1 0 0.840 0.0237 75.8 0.793 0.887 35.4 6.86e-49 0 mg
2 25 0.820 0.0296 119. 0.762 0.879 27.7 9.60e-54 25 mg
3 50 0.838 0.0294 118. 0.779 0.896 28.5 8.15e-55 50 mg
4 100 0.865 0.0289 115. 0.807 0.922 29.9 5.43e-56 100 mg
```

 # i 1 more variable: mean_txt

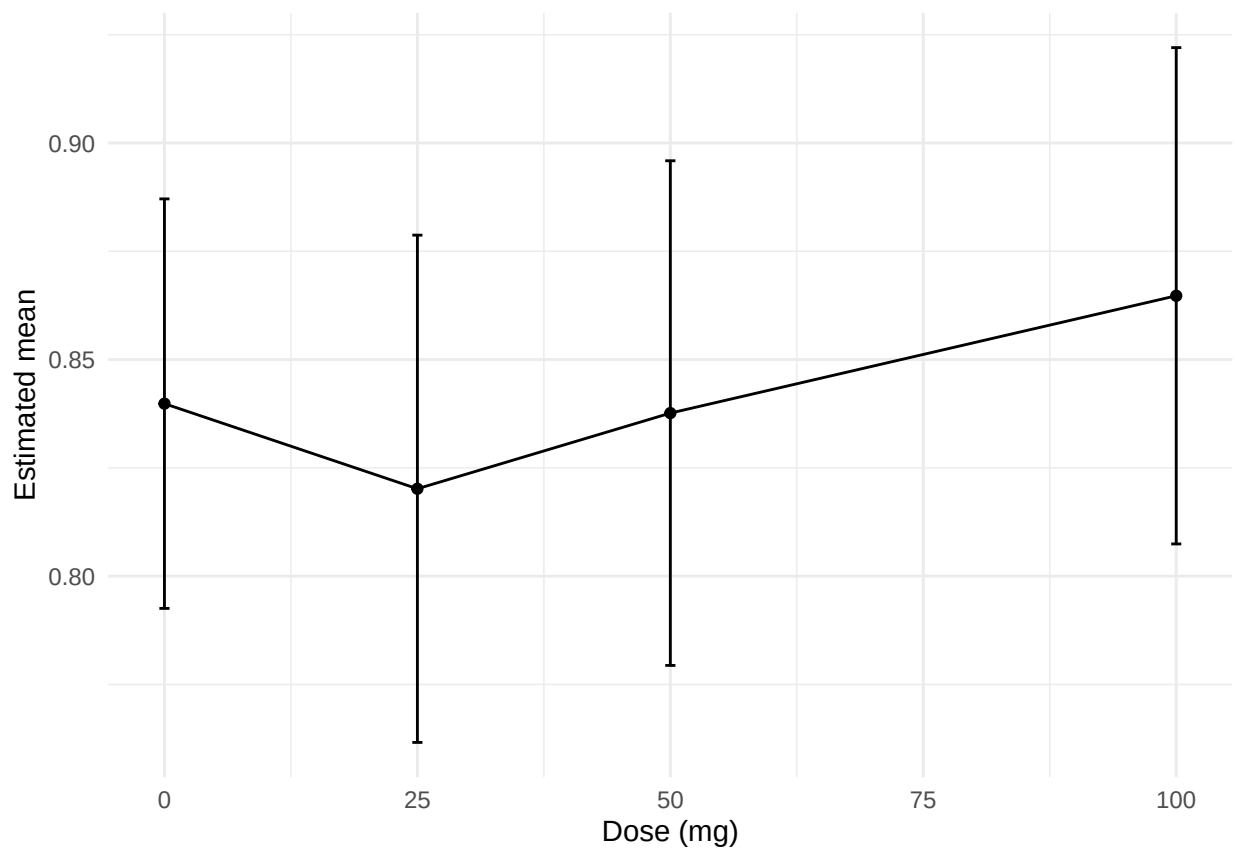
Differences vs 0 mg

A tibble: 3 x 11

```
term contrast null.value estimate std.error df conf.low conf.high
1 trtcd 25 mg vs 0 mg 0 -0.0197 0.0299 112. -0.0911 0.0518
2 trtcd 50 mg vs 0 mg 0 -0.00218 0.0290 109. -0.0716 0.0672
3 trtcd 100 mg vs 0 mg 0 0.0249 0.0286 111. -0.0435 0.0933
```

 # i 3 more variables: statistic , adj.p.value , estimate_txt

Plot



mrdrv

Means by dose

A tibble: 4 x 10

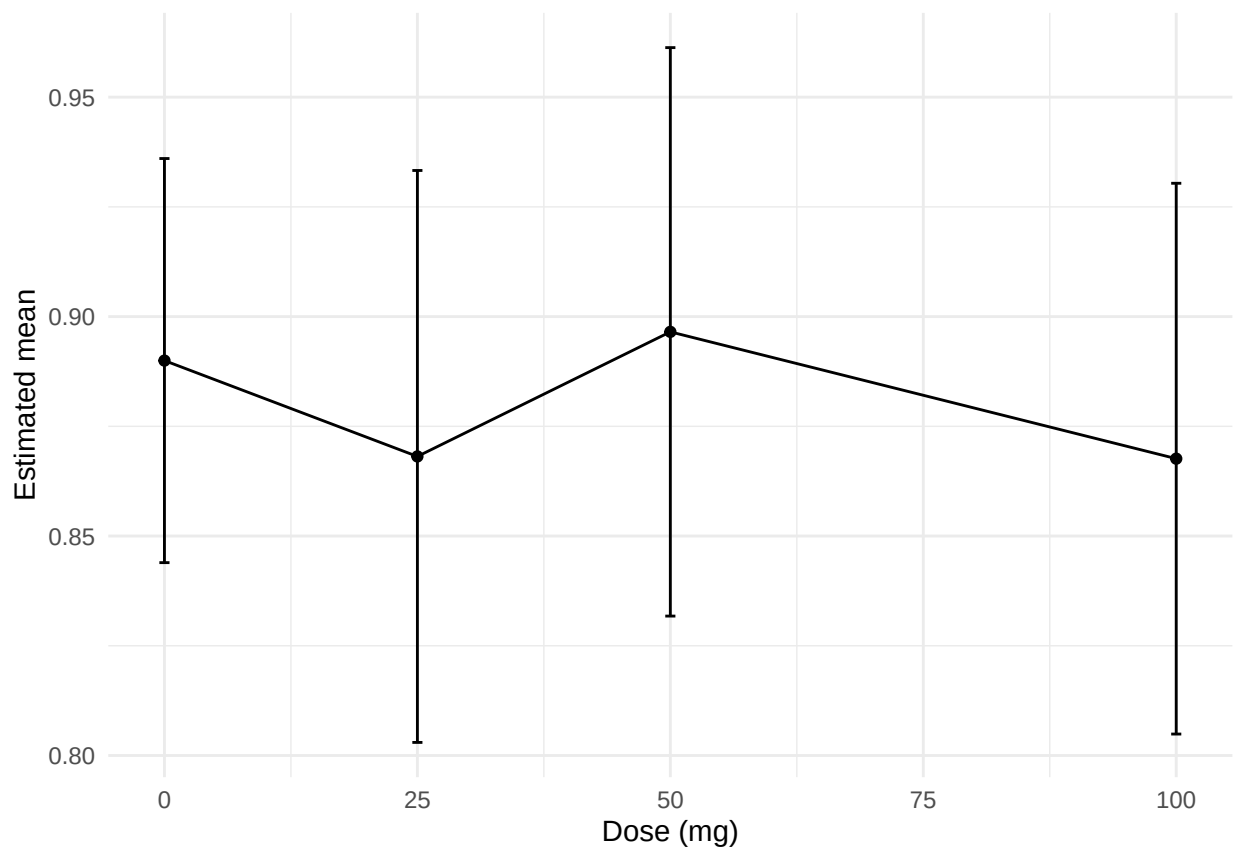
```
trtcd estimate std.error df conf.low conf.high statistic p.value trt
1 0 0.890 0.0232 106. 0.844 0.936 38.3 4.94e-64 0 mg
2 25 0.868 0.0329 134. 0.803 0.933 26.4 7.43e-55 25 mg 3 50 0.897 0.0327 133. 0.832 0.961 27.4 1.34e-56 50
mg 4 100 0.868 0.0317 128. 0.805 0.930 27.4 2.63e-55 100 mg # i 1 more variable: mean_txt
```

Differences vs 0 mg

A tibble: 3 x 11

```
term contrast null.value estimate std.error df conf.low conf.high 1 trtcd 25 mg vs 0 mg 0 -0.0218 0.0406 135.
-0.119 0.0750 2 trtcd 50 mg vs 0 mg 0 0.00655 0.0399 133. -0.0887 0.102 3 trtcd 100 mg vs 0 mg 0 -0.0223
0.0390 136. -0.115 0.0707 # i 3 more variables: statistic , adj.p.value , estimate_txt
```

Plot



mrmts

Means by dose

A tibble: 4 x 10

```
trtcd estimate std.error df conf.low conf.high statistic p.value trt
1 0 0.809 0.0240 74.7 0.761 0.857 33.7 6.28e-47 0 mg
2 25 0.786 0.0298 117. 0.727 0.845 26.4 2.91e-51 25 mg
3 50 0.801 0.0296 117. 0.742 0.859 27.0 3.65e-52 50 mg
4 100 0.827 0.0291 113. 0.769 0.885 28.4 2.31e-53 100 mg
```

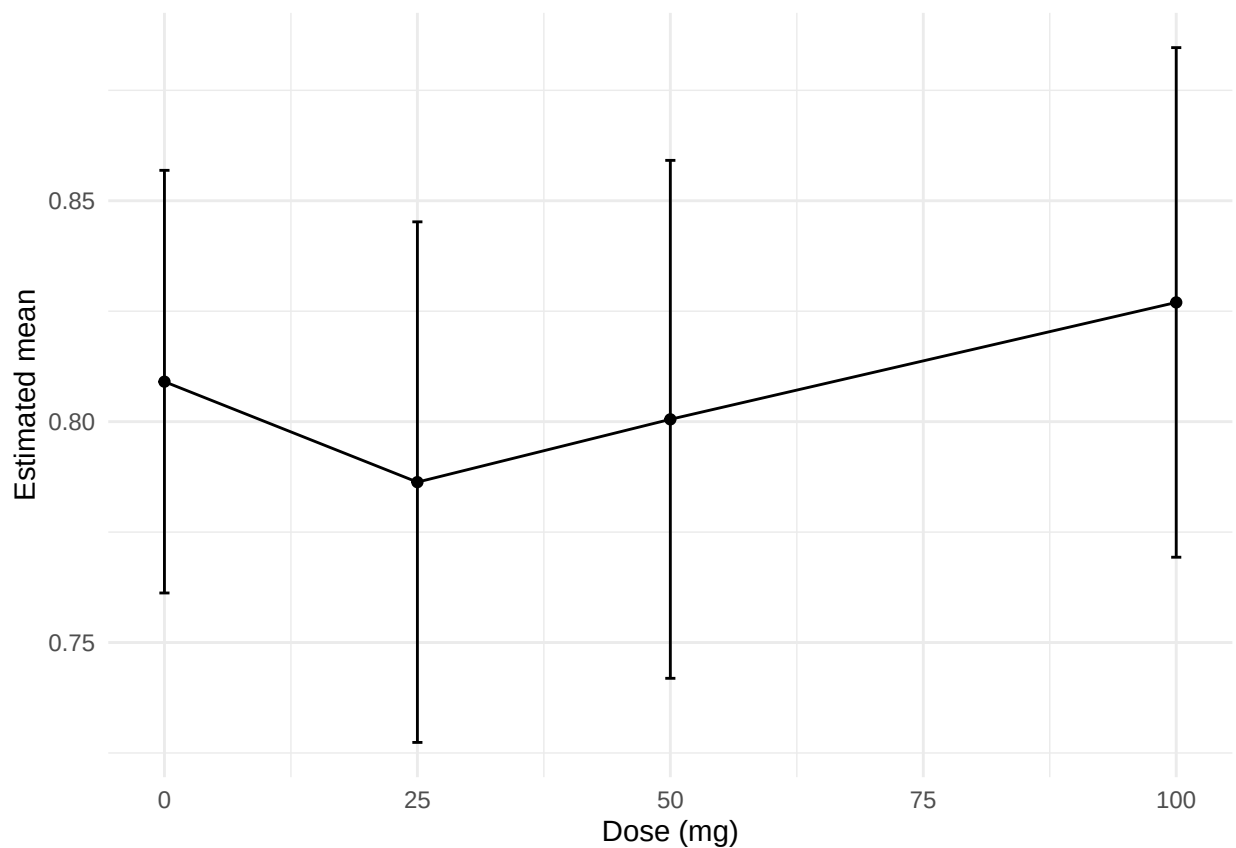
i 1 more variable: mean_txt

Differences vs 0 mg

A tibble: 3 x 11

```
term contrast null.value estimate std.error df conf.low conf.high 1 trtcd 25 mg vs 0 mg 0 -0.0227 0.0297 111.
-0.0939 0.0484 2 trtcd 50 mg vs 0 mg 0 -0.00853 0.0288 108. -0.0775 0.0605 3 trtcd 100 mg vs 0 mg 0 0.0179
0.0284 110. -0.0501 0.0860 # i 3 more variables: statistic , adj.p.value , estimate_txt
```

Plot



mrdiv

Means by dose

A tibble: 4 x 10

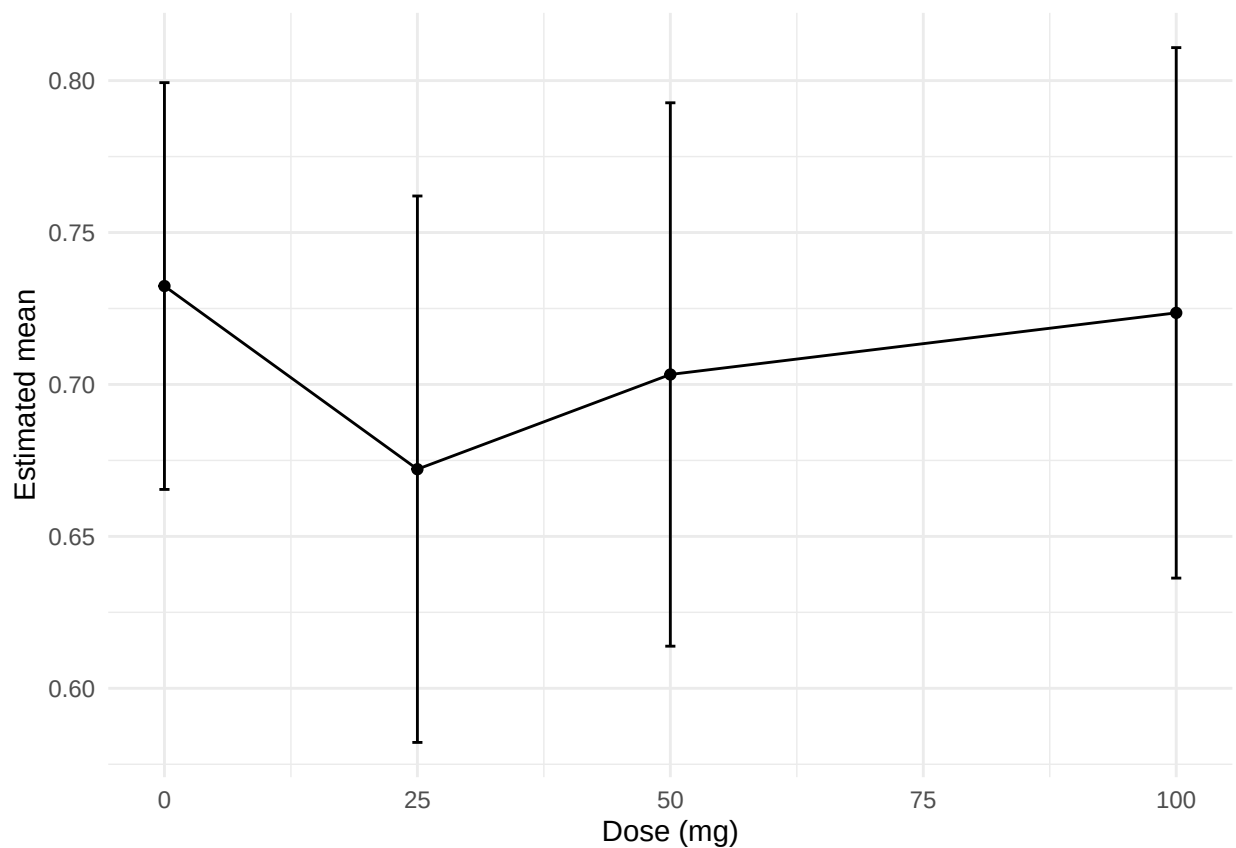
```
trtcd estimate std.error df conf.low conf.high statistic p.value trt
1 0 0.732 0.0337 93.8 0.665 0.799 21.7 2.27e-38 0 mg
2 25 0.672 0.0455 132. 0.582 0.762 14.8 8.25e-30 25 mg
3 50 0.703 0.0452 132. 0.614 0.793 15.6 1.17e-31 50
mg 4 100 0.724 0.0441 128. 0.636 0.811 16.4 2.83e-33 100 mg # i 1 more variable: mean_txt
```

Differences vs 0 mg

A tibble: 3 x 11

```
term contrast null.value estimate std.error df conf.low conf.high 1 trtcd 25 mg vs 0 mg 0 -0.0603 0.0522 124.
-0.185 0.0645 2 trtcd 50 mg vs 0 mg 0 -0.0291 0.0510 121. -0.151 0.0928 3 trtcd 100 mg vs 0 mg 0 -0.00883
0.0501 123. -0.128 0.111 # i 3 more variables: statistic , adj.p.value , estimate_txt
```

Plot



mrt1ca

Means by dose

A tibble: 4 x 10

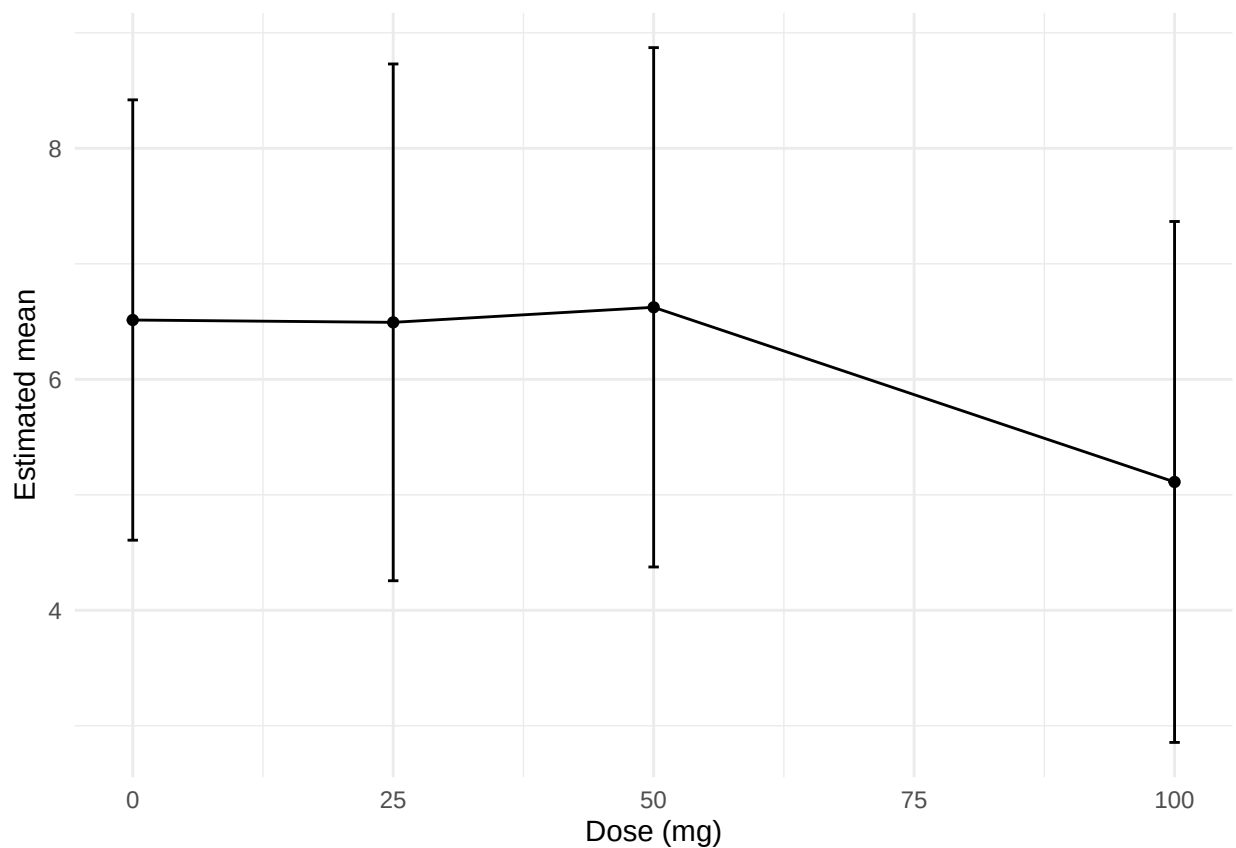
```
trtcd estimate std.error df conf.low conf.high statistic p.value trt
1 0 6.51 0.957 75.4 4.61 8.42 6.81 2.10e-9 0 mg 2 25 6.49 1.13 123. 4.26 8.73 5.74 6.79e-8 25 mg 3 50 6.62
1.14 124. 4.37 8.87 5.83 4.47e-8 50 mg 4 100 5.11 1.14 125. 2.86 7.37 4.49 1.63e-5 100 ~ # i 1 more variable:
mean_txt
```

Differences vs 0 mg

A tibble: 3 x 11

```
term contrast null.value estimate std.error df conf.low conf.high 1 trtcd 25 mg vs 0 mg 0 -0.0202 0.977 153.
-2.35 2.31 2 trtcd 50 mg vs 0 mg 0 0.110 0.980 154. -2.23 2.45 3 trtcd 100 mg vs 0 mg 0 -1.40 0.988 155. -3.76
0.952 # i 3 more variables: statistic , adj.p.value , estimate_txt
```

Plot



mrt2ps

Means by dose

A tibble: 4 x 10

```
trtcd estimate std.error df conf.low conf.high statistic p.value trt
1 0 19.7 2.95 83.4 13.8 25.5 6.67 2.59e-9 0 mg 2 25 19.4 3.63 142. 12.3 26.6 5.36 3.35e-7 25 mg 3 50 19.0
3.65 143. 11.8 26.2 5.21 6.45e-7 50 mg 4 100 17.9 3.66 143. 10.7 25.1 4.89 2.72e-6 100 ~ # i 1 more variable:
mean_txt
```

Differences vs 0 mg

A tibble: 3 x 11

```
term contrast null.value estimate std.error df conf.low conf.high 1 trtcd 25 mg vs 0 mg 0 -0.230 3.45 158.
-8.46 8.00 2 trtcd 50 mg vs 0 mg 0 -0.647 3.46 158. -8.90 7.61 3 trtcd 100 mg vs 0 mg 0 -1.77 3.49 160. -10.1
6.54 # i 3 more variables: statistic , adj.p.value , estimate_txt
```

Plot

